

An illustration in a dark teal and black color palette. On the right, a figure in a dark hooded robe wears a white mask with large, dark, oval eyes and a wide, open mouth, resembling a ghost or a 'Scream' mask. The figure is looking down at a laptop. In the background on the left, there are two overlapping browser window icons. The top window shows two white ghost-like figures with large eyes and open mouths. The bottom window shows a single white ghost-like figure with a large eye and open mouth. The background is dark with some faint orange lines.

Ghost Calls: Abusing Web Conferencing for Covert Command & Control

Adam Crosser

Introduction



Adam Crosser



Praetorian

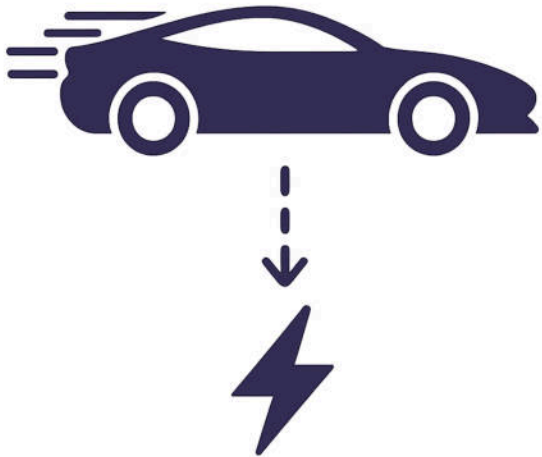
LinkedIn: <https://www.linkedin.com/in/adam-crosser-366263265>

X: <https://x.com/UNC1739>

Types of Command-and-Control Channels

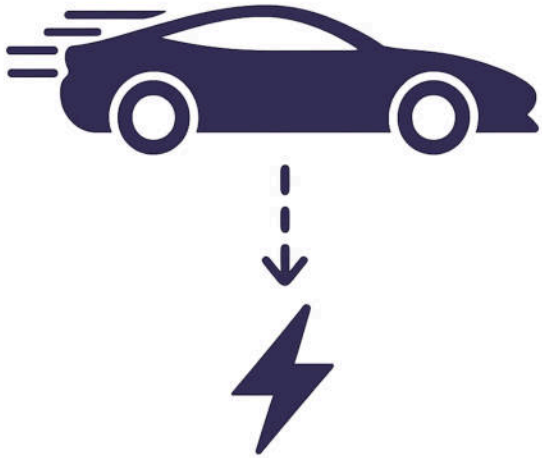
Types of Command-and-Control Channels

SHORT

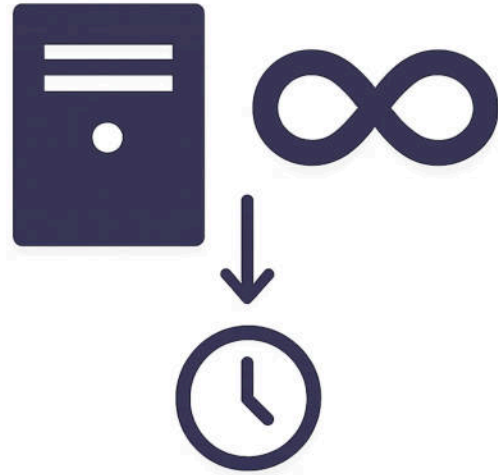


Types of Command-and-Control Channels

SHORT

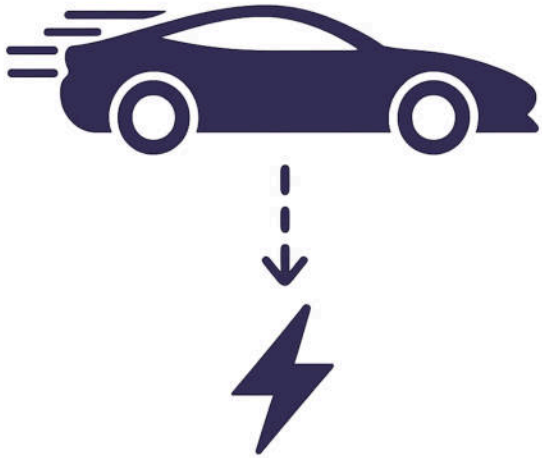


LONG

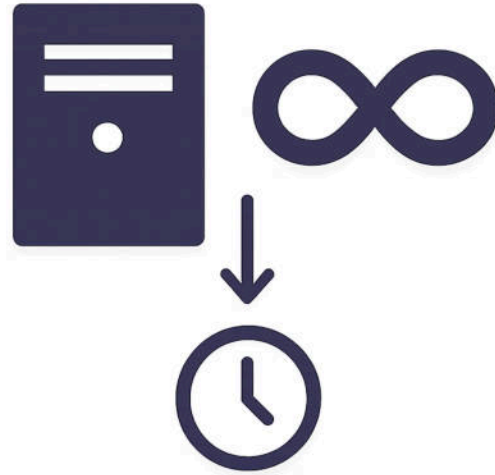


Types of Command-and-Control Channels

SHORT



LONG

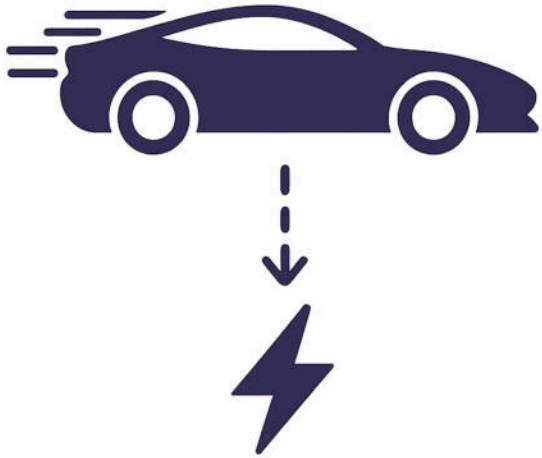


BACKUP

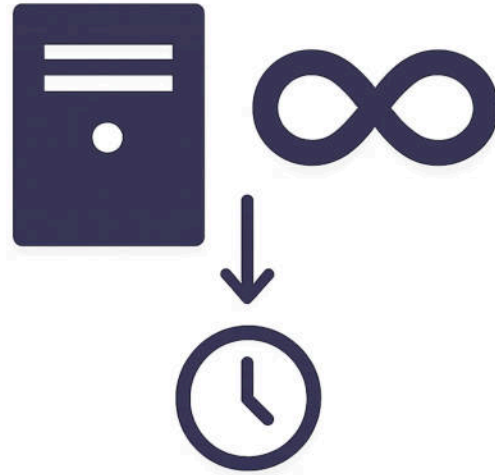


Types of Command-and-Control Channels

SHORT



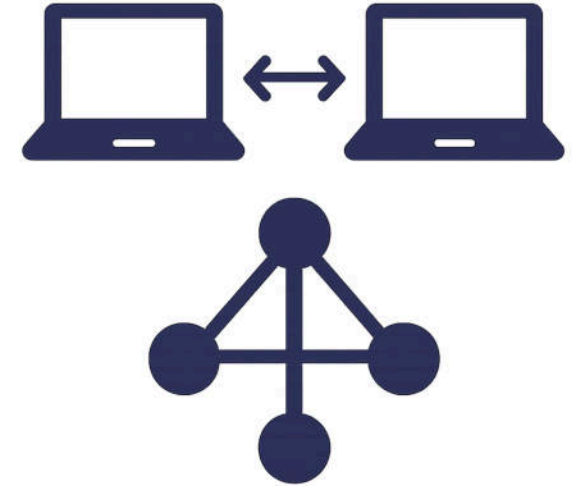
LONG



BACKUP



P2P



BRAINSTORMING SOLUTIONS

**CLOUD
STORAGE**



**DNS OVER
HTTPS**

**EMAIL
MESSAGING**



B64

EMAIL



Ideal Short-Term Command and Control

Ideal Short-Term Command and Control

LATENCY



Ideal Short-Term Command and Control

LATENCY

THROUGHPUT



Ideal Short-Term Command and Control

LATENCY



THROUGHPUT



REACH



Ideal Short-Term Command and Control

LATENCY



THROUGHPUT



REACH



TRUST

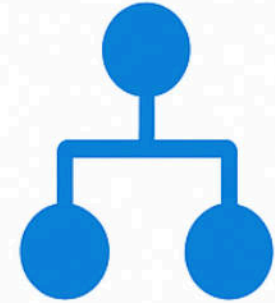


Selection Criteria

- Focused on services egressing from user devices
- Must be broadly used across enterprise roles
- Applicable to non-technical departments (e.g., HR, sales)
- Protocols favored by technical users were excluded
- Thought through common workflows and use-cases



HTTP



DNS



SMB



RDP

DNS over HTTP (DoH)



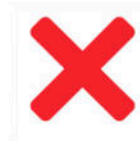
DNS over HTTPS

- ✓ LATENCY
- ✗ THROUGHPUT
- REACH
- ✓ TRUST

Cloud File Storage



box



LATENCY



THROUGHPUT



REACH



TRUST

Attacker Virtual Machine with Classified Domain



LATENCY



THROUGHPUT



REACH



TRUST

Email and Messaging Applications



LATENCY



THROUGHPUT



REACH



TRUST

Web Conferencing



- ✓ **LATENCY**
- ✓ **THROUGHPUT**
- ✓ **REACH**
- ✓ **TRUST**

Microsoft Teams Split Tunneling Guidelines

Configure split-tunnel VPN

We recommend that you provide an alternate path for Teams traffic that bypasses the virtual private network (VPN), commonly known as [split-tunnel VPN](#). Split tunneling means that traffic for Microsoft 365 or Office 365 doesn't go through the VPN but instead goes directly to Microsoft 365 or Office 365. Bypassing your VPN has a positive impact on Teams quality, and it reduces load from the VPN devices and the organization's network. To implement a split-tunnel VPN, work with your VPN vendor.

Other reasons why we recommend bypassing the VPN:

- VPNs are typically not designed or configured to support real-time media.
- Some VPNs might also not support UDP (which is required for Teams).
- VPNs also introduce an extra layer of encryption on top of media traffic that's already encrypted.
- Connectivity to Teams might not be efficient due to hair-pinning traffic through a VPN device.
- Traffic might be routed to a service front door location that is further away from the end user, introducing extra latency and jitter.



<https://learn.microsoft.com/en-us/microsoftteams/prepare-network>

Microsoft Teams TLS Inspection

Not using a proxy server is recommended



Many organizations utilize proxy servers today within their network. As Microsoft Teams and Skype for Business traffic is already encrypted, passing this traffic through a proxy server doesn't make the traffic any more secure.

Proxies can cause issues too. Performance-related problems can be introduced to the environment through latency and packet loss by attempting to route Teams traffic through a proxy server. This can be caused by the proxy being unable to handle the amount of traffic passing through it, or by incorrectly routing the traffic to a Microsoft network service front door location that is further away from the end user.

Issues such as these will result in a negative experience within Teams and Skype for Business.

We recommend that Teams traffic bypasses proxy server infrastructure, including SSL inspection. You may wish to achieve this by putting Teams Phones and Meeting Room devices on their own VLAN and providing them with Internet access.

<https://learn.microsoft.com/en-us/microsoftteams/proxy-servers-for-skype-for-business-online>

Zoom Split Tunneling Recommendations

VPN Split Tunneling Recommendations



English (Original)



2025-03-06 16:55:57



Copy Permalink

Virtual Private Network (VPN) services are crucial to securing data accessed by users working from remote locations.

One of the biggest challenges Zoom customers experience is related to not allowing our real-time media services over UDP 8801-8810 to split tunnel. Not allowing split tunneling for UDP 8801-8810 and TCP 443 to Zoom resources, does cause customers to experience significant additional load on their corporate internet connections due to the Zoom traffic having to enter the corporate network, only to exit again to the Zoom cloud for real-time meeting termination. This also places a significant amount of burden on VPN concentrators and in many cases can cause overloading and congestion of this infrastructure.

https://support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0065998

Zoom TLS Inspection Recommendations



Proxy server

We support HTTPS/SSL proxy servers via port 443 for Zoom traffic.

Note: This does not apply to the Zoom Phone service.

Zoom automatically detects your proxy settings. In some instances, you may be prompted to enter the proxy username/password.

Note: We recommend allowing **zoom.us** and ***.zoom.us** from proxy or SSL inspection.

https://support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0060548

Quick Disclaimer

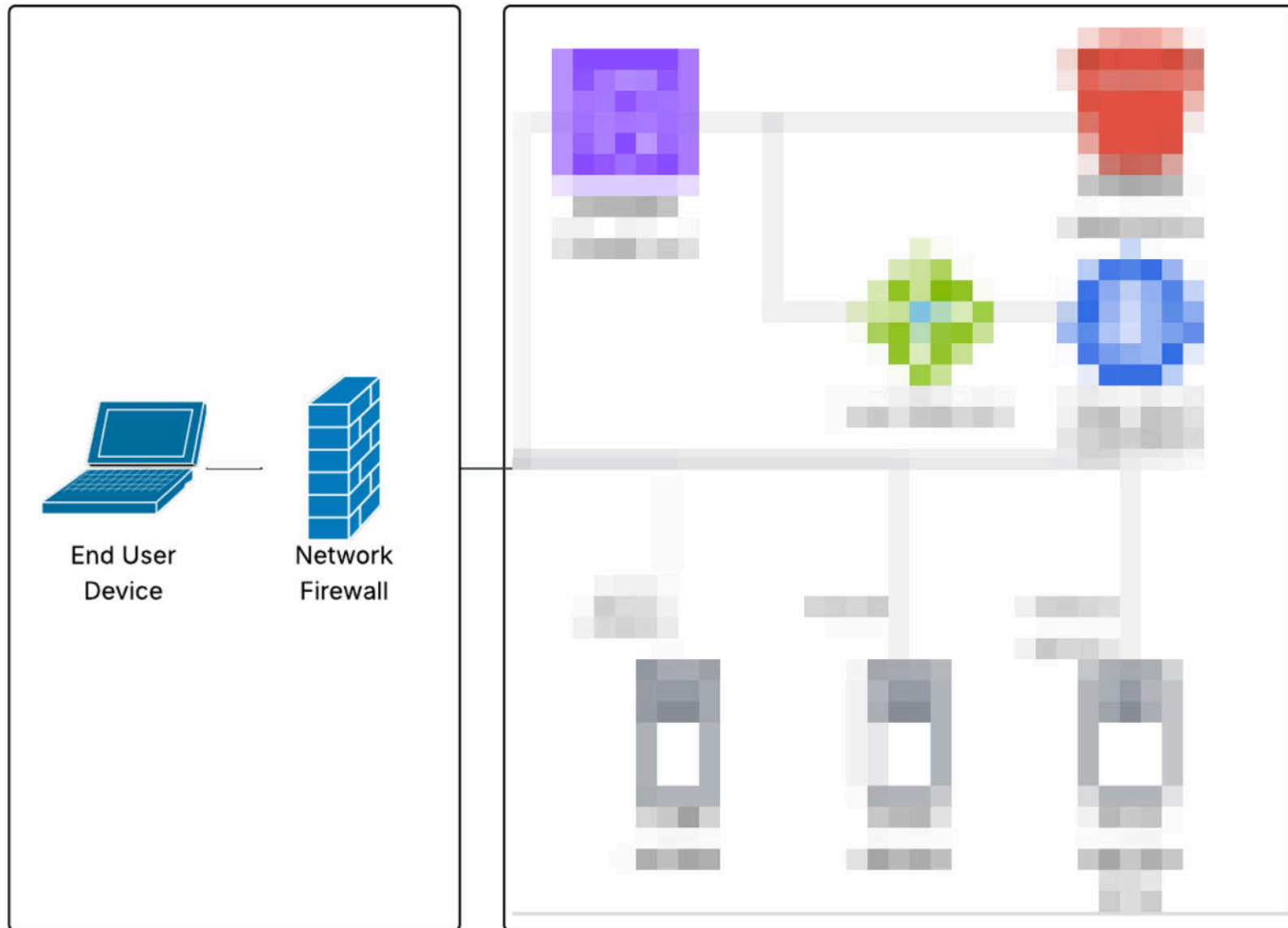
- Providers aren't being malicious
- Performance is the main design driver
- Latency must be minimized for app reliability
- These configs are often intentional not careless
- Inspection or routing can overwhelm systems



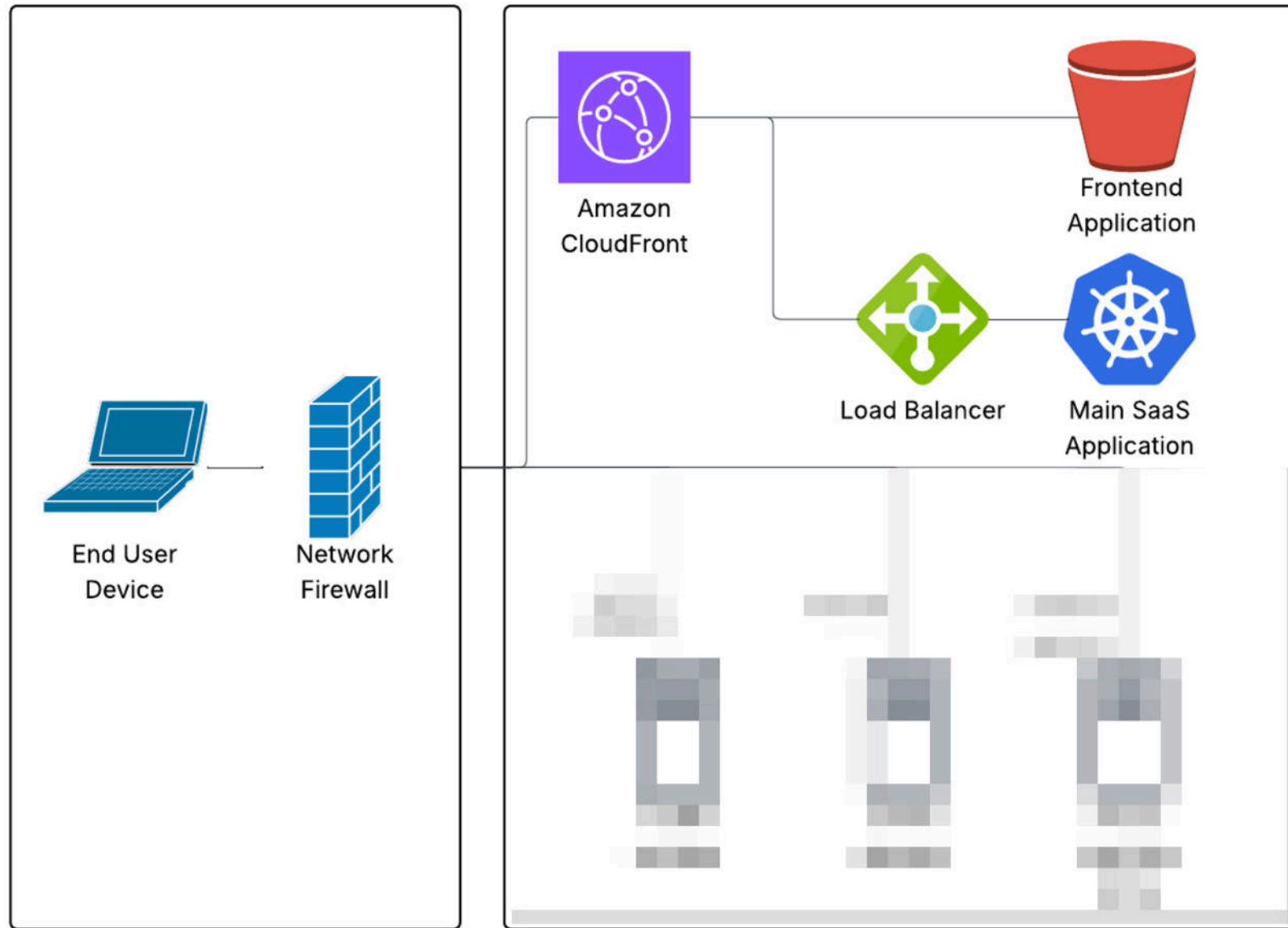
HOW IT WORKS



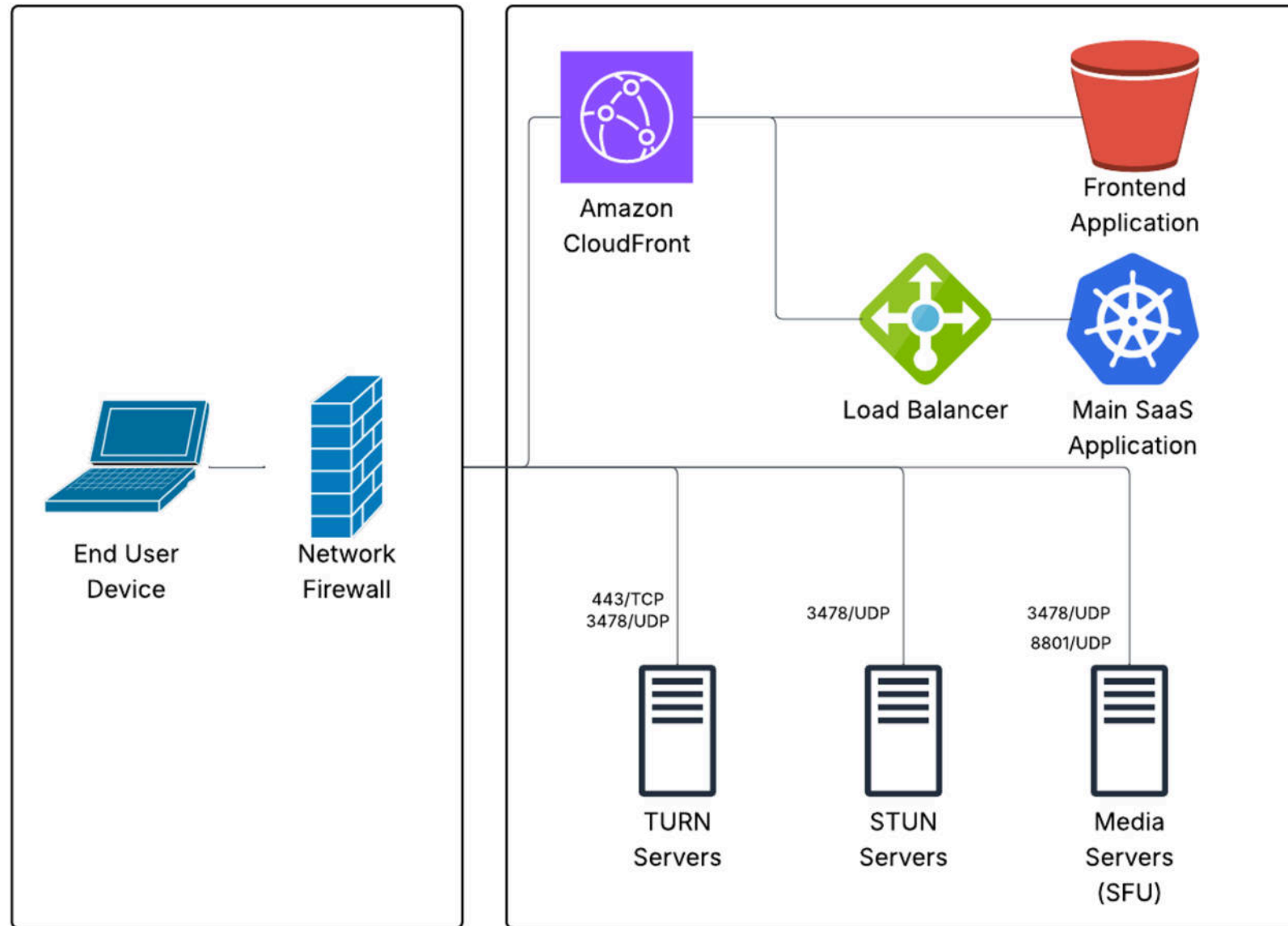
General Web Conferencing Architecture



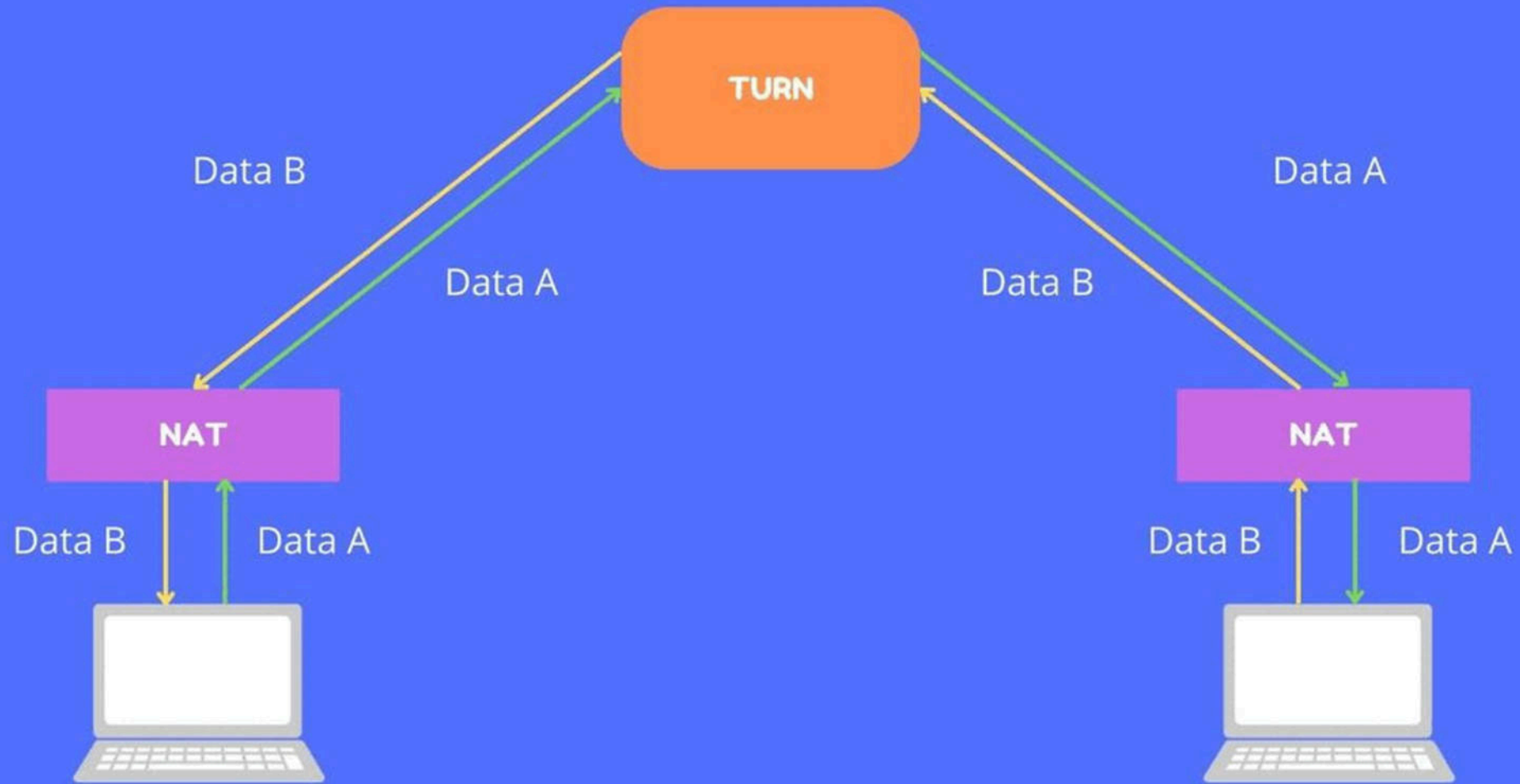
General Web Conferencing Architecture



General Web Conferencing Architecture



What is TURN?



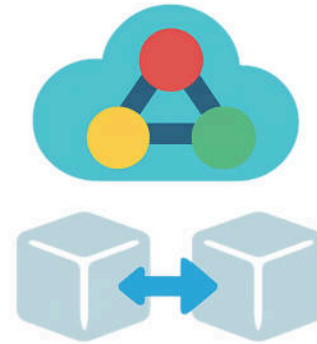
WebRTC Core Protocols



DTLS



SCTP

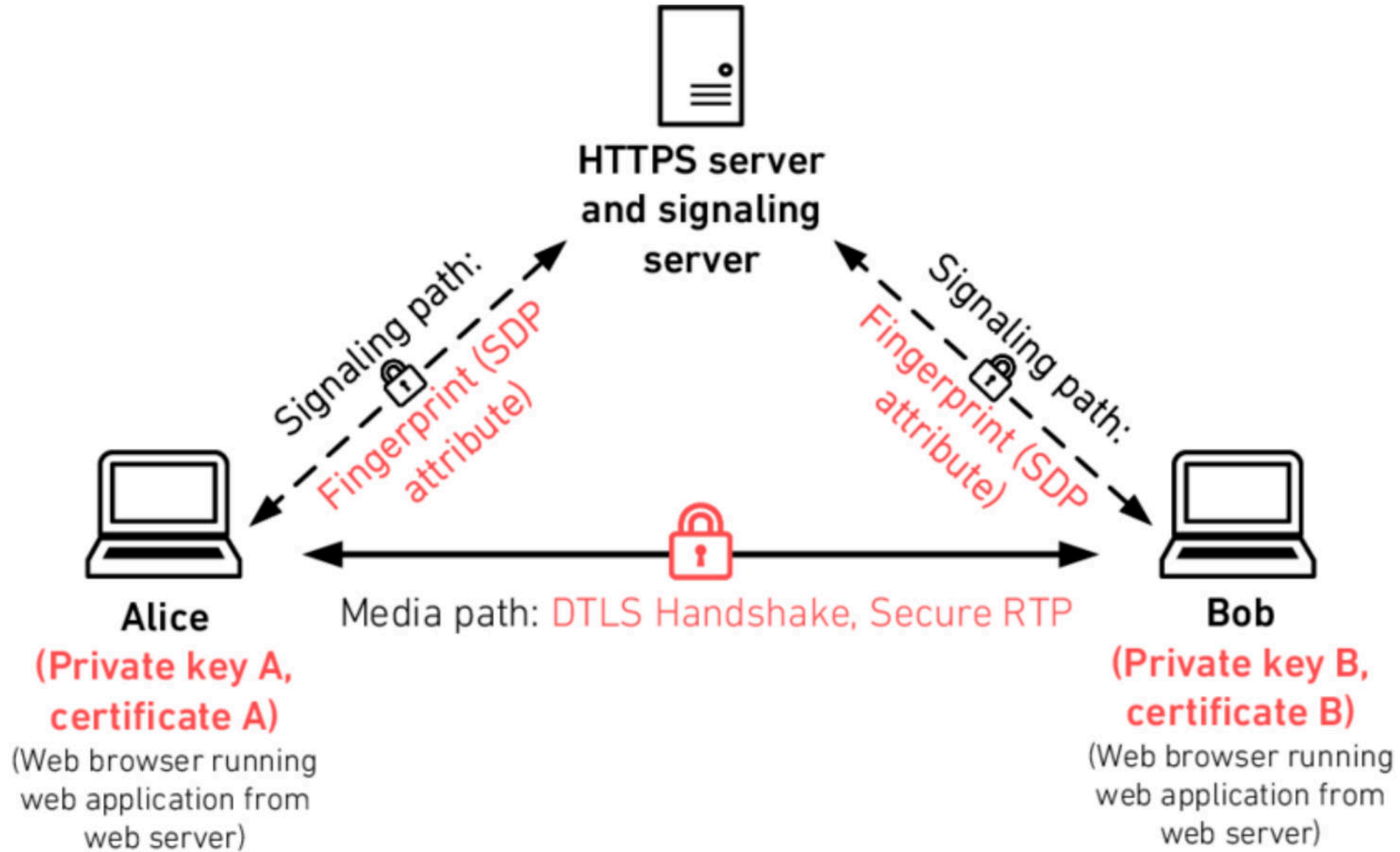


ICE



SRTP

WebRTC Handshake Process



Reverse Engineering Zoom

```
> Internet Protocol Version 4, Src: 192.168.1.41, Dst: 170.114.164.95
> User Datagram Protocol, Src Port: 61029, Dst Port: 8801
✓ Zoom SFU Encapsulation
  Type: 5
  Sequence number: 1278
  Direction: 0 (to Zoom)
✓ Zoom Media Encapsulation
  Type: 16 (Video)
  Sequence number: 1261
  Timestamp: 106179922
  Frame number: 57
  Packets in frame: 2
✓ Real-Time Transport Protocol
  > [Stream setup by DECODE AS (frame 28373)]
  10.. .... = Version: RFC 1889 Version (2)
  ..0. .... = Padding: False
  ...1 .... = Extension: True
  .... 0000 = Contributing source identifiers count: 0
  1... .... = Marker: True
  Payload type: DynamicRTP-Type-98 (98)
  Sequence number: 24484
  [Extended sequence number: 90020]
  Timestamp: 894589134
  [Extended timestamp: 5189556430]
  Synchronization Source identifier: 0x01000401 (16778241)
  Defined by profile: RFC 5285 One-Byte Header Extensions (0xbede)
  Extension length: 5
  > Header extensions
  Payload [...]: 1c40736b27a5415cf9715dd657876f8c59f14a70c4c6878987c74f26b8123f633690b6ef5ccee1e88f5932228eadc93eefe91c9f2:
```


Reverse Engineering Zoom

- > Internet Protocol Version 4, Src: 192.168.1.41, Dst: 170.114.164.95
- > User Datagram Protocol, Src Port: 61029, Dst Port: 8801
- ✓ Zoom SFU Encapsulation

Reverse Engineering Zoom

- > User Datagram Protocol, Src Port: 61029, Dst Port: 8801
- ✓ Zoom SFU Encapsulation
 - Type: 5
 - Sequence number: 1278
 - Direction: 0 (to Zoom)
- ✓ Zoom Media Encapsulation
 - Type: 16 (Video)
 - Sequence number: 1261
 - Timestamp: 106179922
 - Frame number: 57
 - Packets in frame: 2
- ✓ Real-Time Transport Protocol
 - > [Stream setup by DECODE AS (frame 28373)]

Building on Existing Work

Enabling Passive Measurement of Zoom
Performance in Production Networks



<https://dl.acm.org/doi/pdf/10.1145/3517745.3561414>

Custom Wireshark Analyzer for Zoom Desktop
Media Traffic



<https://github.com/Princeton-Cabernet/zoom-analysis>

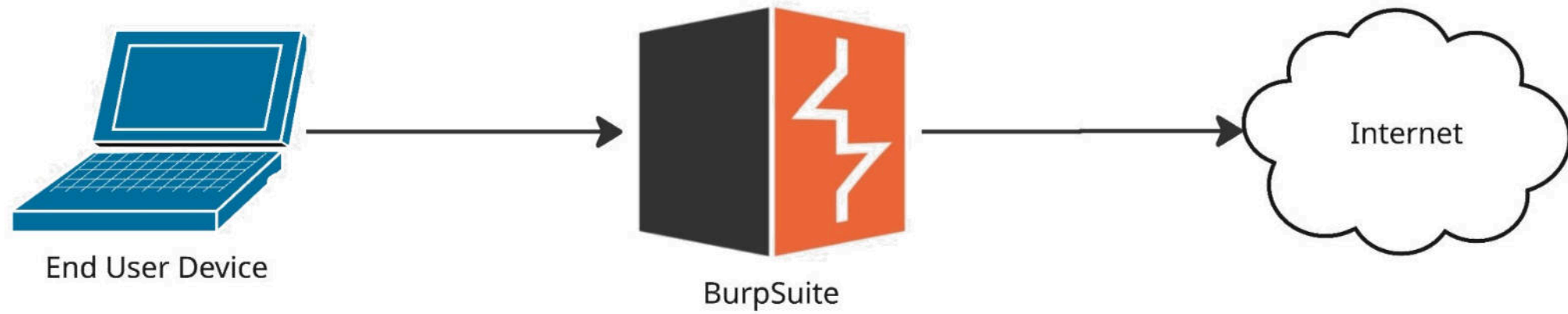
Reverse Engineering Google Meet

```
> Frame 4296: 160 bytes on wire (1280 bits), 160 bytes captured (1280 bits)
> Ethernet II, Src: Apple_d5:f9:5f (14:7d:da:d5:f9:5f), Dst: zte_4c:ac:24 (20:08:89:4c:ac:24)
> Internet Protocol Version 4, Src: 192.168.1.43, Dst: 74.125.250.251
> User Datagram Protocol, Src Port: 63070, Dst Port: 3478
✓ Real-Time Transport Protocol
  > [Stream setup by DTLS-SRTP (frame 2963)]
    10.. .... = Version: RFC 1889 Version (2)
    ..0. .... = Padding: False
    ...1 .... = Extension: True
    .... 0000 = Contributing source identifiers count: 0
    0... .... = Marker: False
    Payload type: Unassigned (63)
    Sequence number: 24725
    [Extended sequence number: 90261]
    Timestamp: 345165098
    [Extended timestamp: 4640132394]
    Synchronization Source identifier: 0xa11f30c7 (2703175879)
    Defined by profile: RFC 5285 One-Byte Header Extensions (0xbede)
    Extension length: 3
  ✓ Header extensions
    > RFC 5285 Header Extension (One-Byte Header)
    > RFC 5285 Header Extension (One-Byte Header)
    > RFC 5285 Header Extension (One-Byte Header)
    SRTP Encrypted Payload: 56046ee649b15872c7d5a1f0b3604bf7ee71d42d8d55062dc3c6a639ae063054d04ea8469f2495cf5c34
    SRTP Auth Tag: 175d3f6ef64838a438b484a7dee2dbbc
```


Reverse Engineering Google Meet

- > Internet Protocol Version 4, Src: 192.168.1.43, Dst: 74.125.250.251
- > User Datagram Protocol, Src Port: 63070, Dst Port: 3478
- ✓ Real-Time Transport Protocol
 - > [Stream setup by DTLS-SRTP (frame 2963)]
 - 10.. = Version: RFC 1889 Version (2)
 - ..0. = Padding: False
 - ...1 = Extension: True
 - 0000 = Contributing source identifiers count: 0

Case Study in Egress Resilience



Example Zoom Desktop Egress Attempts



Zone Controller
Primary



MultiMedia Router

Example Zoom Desktop Egress Attempts



Custom Protocol over TLS on 443/TCP



Zone Controller
Primary



MultiMedia Router

Example Zoom Desktop Egress Attempts



Custom Protocol  TLS on 443/TCP



Zone Controller
Primary



MultiMedia Router

Example Zoom Desktop Egress Attempts



WebSockets over HTTPS on 443/TCP



Zone Controller
Primary



MultiMedia Router

Example Zoom Desktop Egress Attempts



WebSockets over HTTPS on 443/TCP

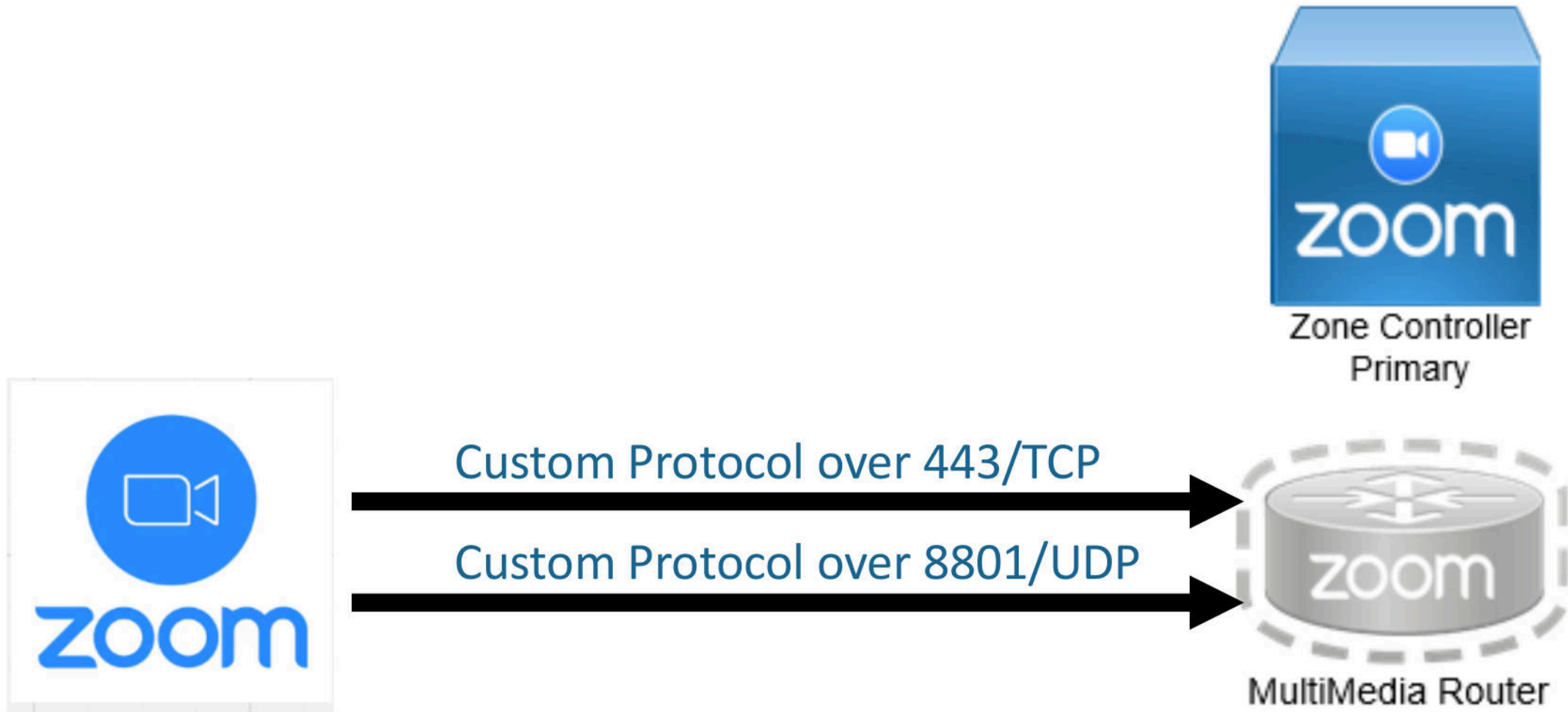


Zone Controller
Primary

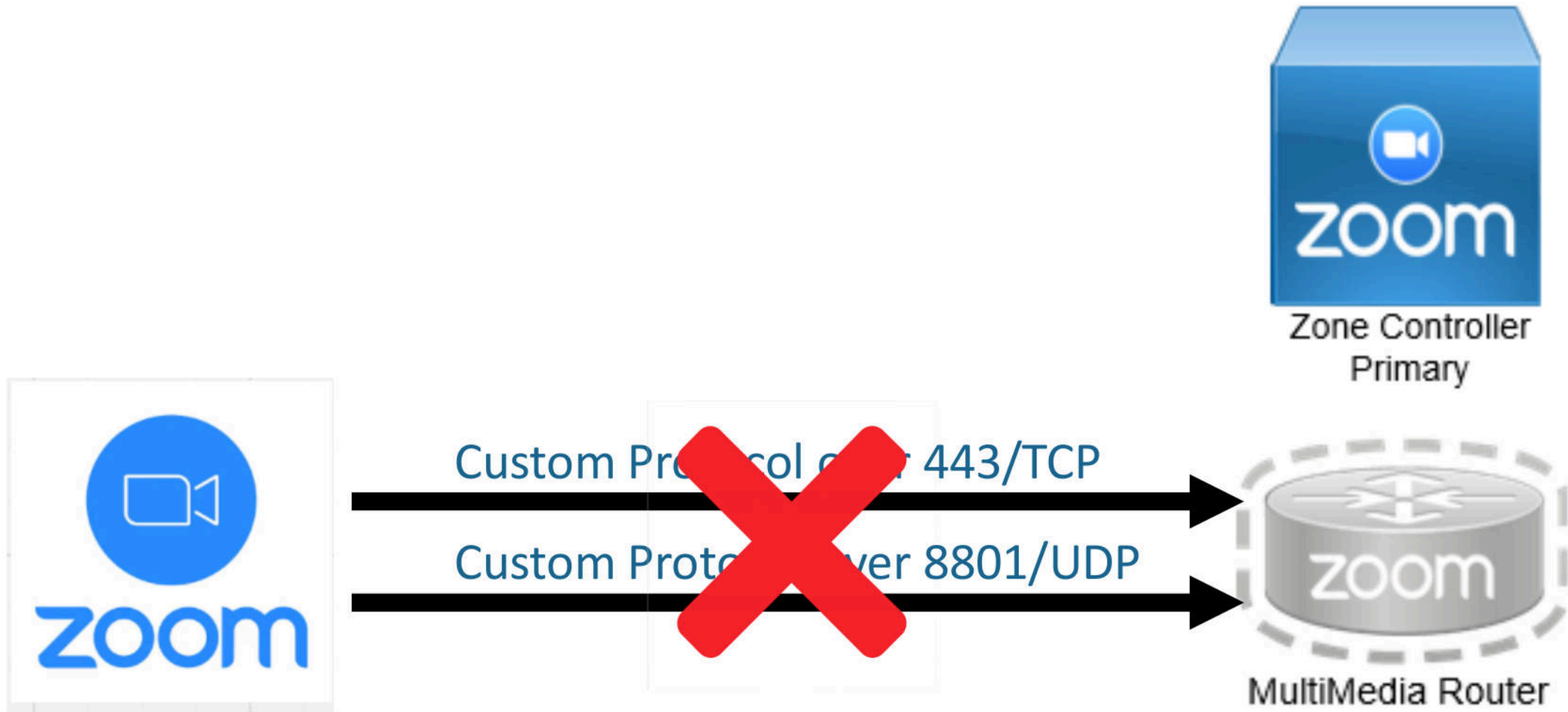


MultiMedia Router

Example Zoom Desktop Egress Attempts



Example Zoom Desktop Egress Attempts



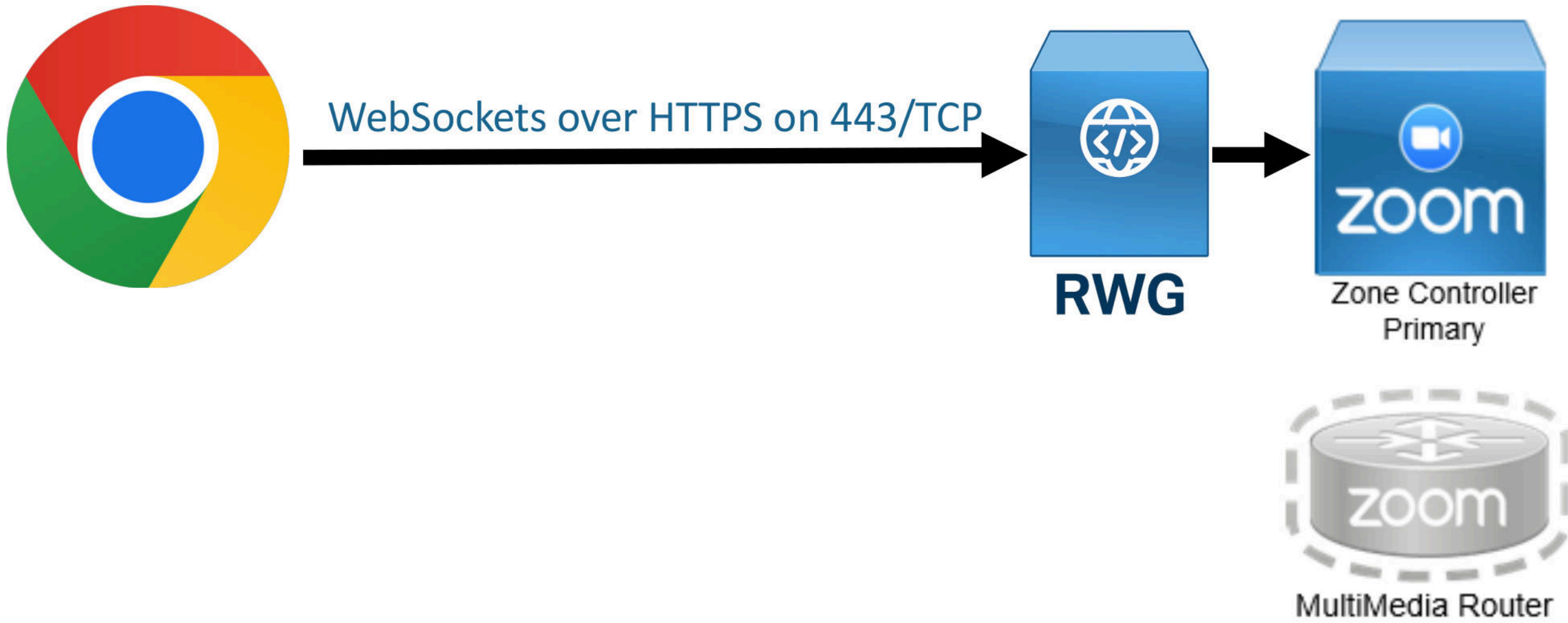
Example Zoom Desktop Egress Attempts



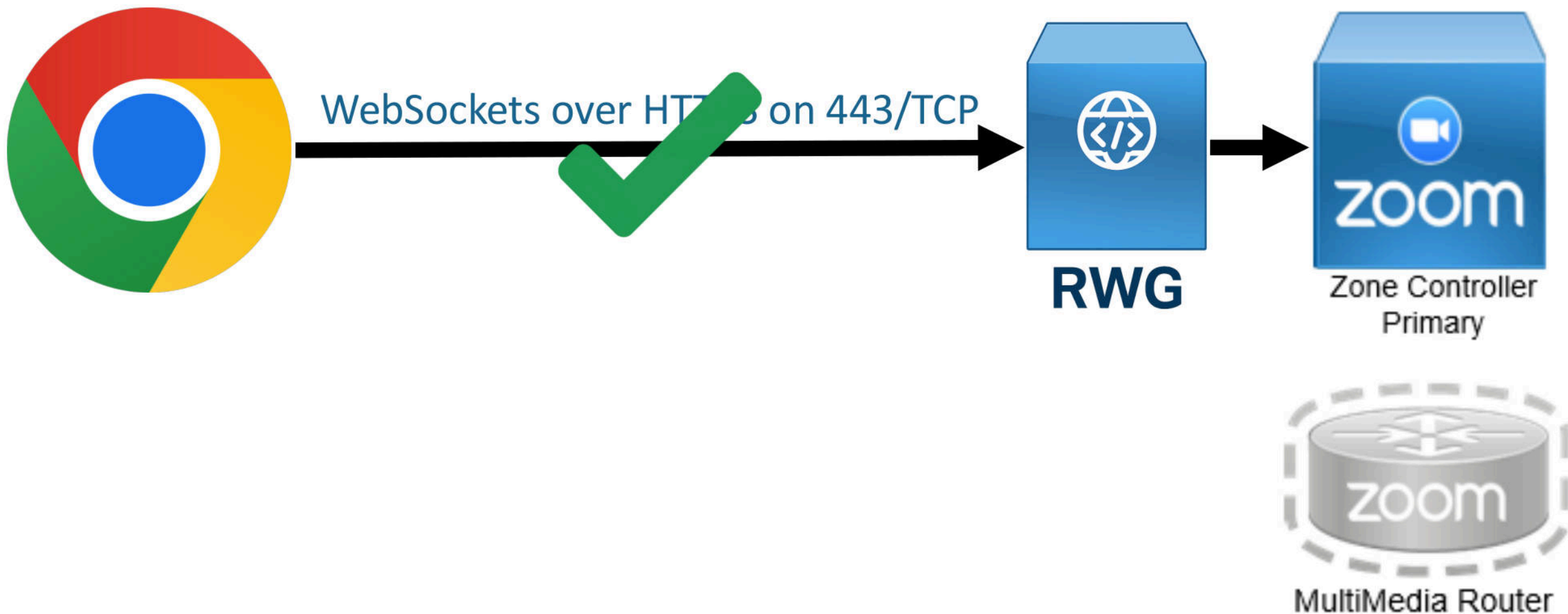
Example Zoom Desktop Egress Attempts



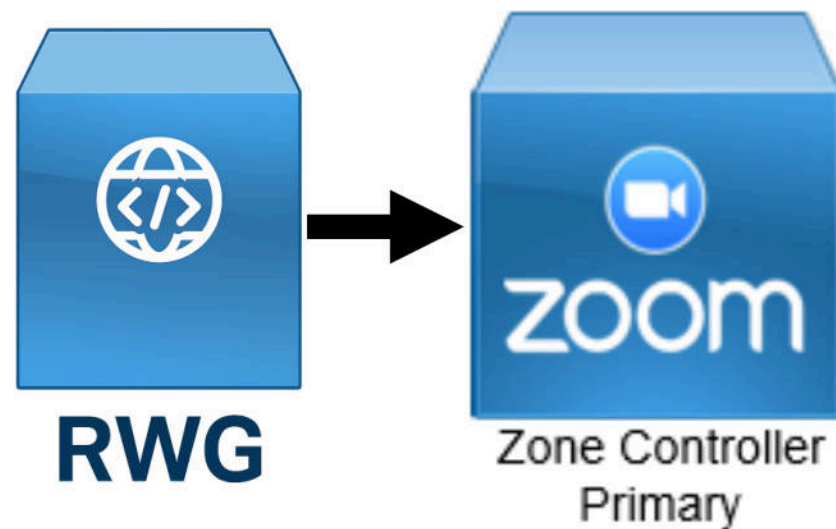
Example Zoom Web Client Egress Attempts



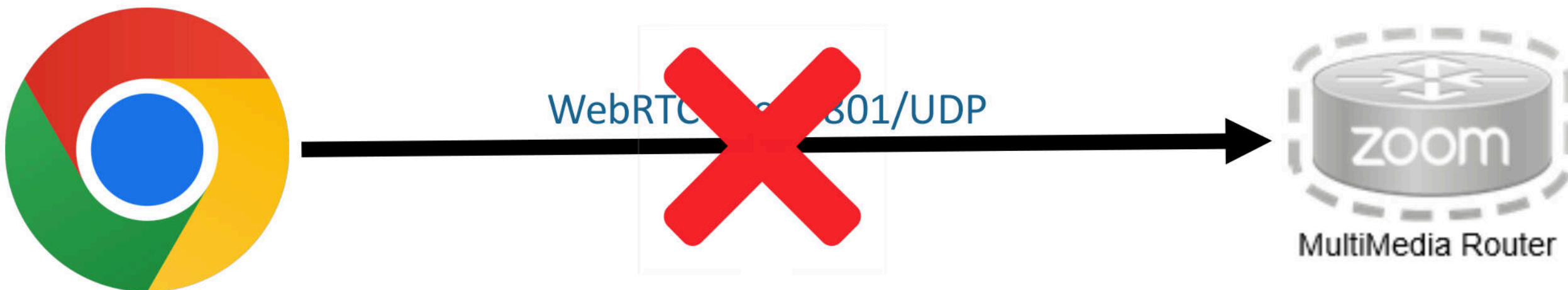
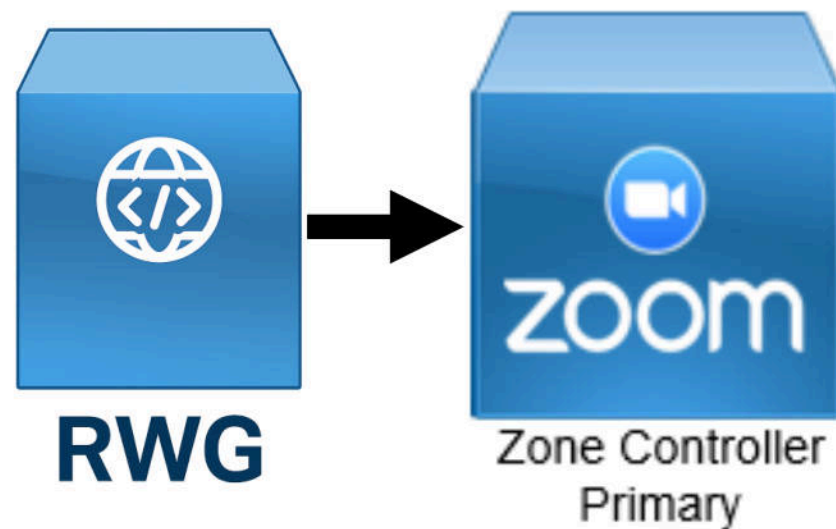
Example Zoom Web Client Egress Attempts



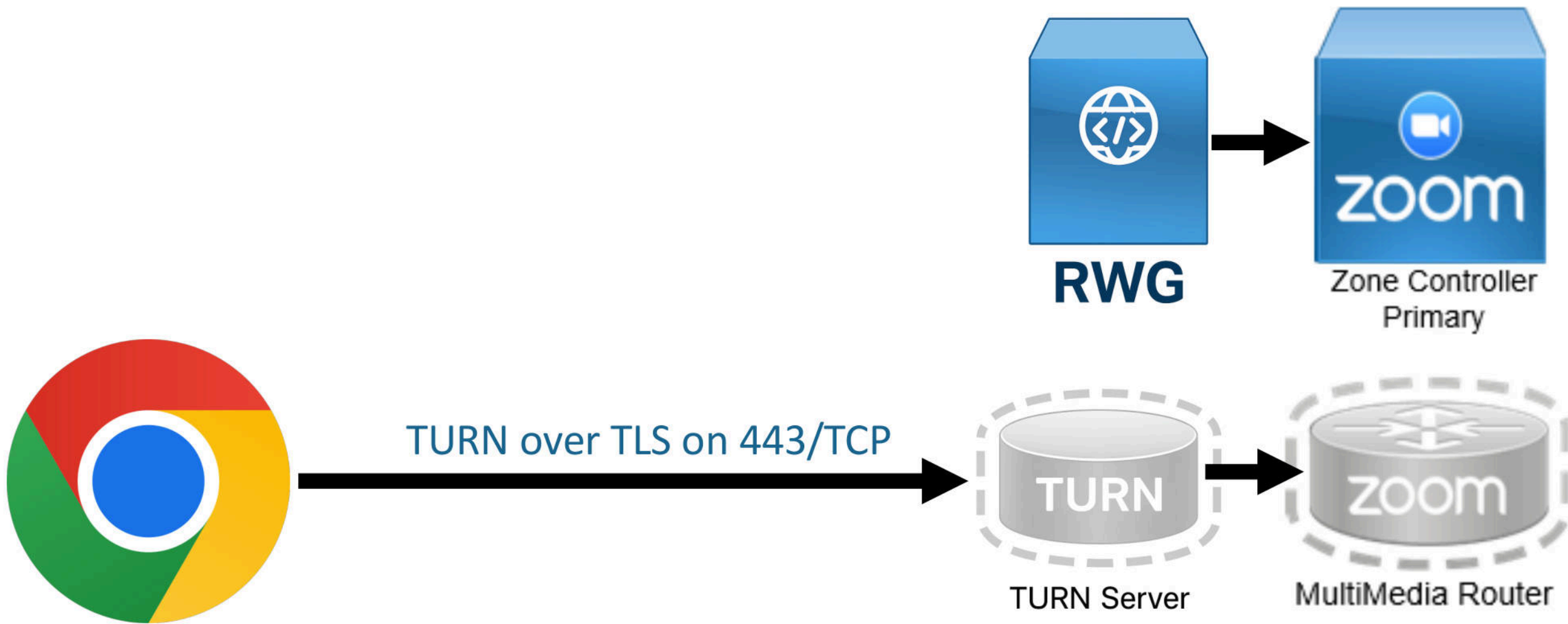
Example Zoom Web Client Egress Attempts



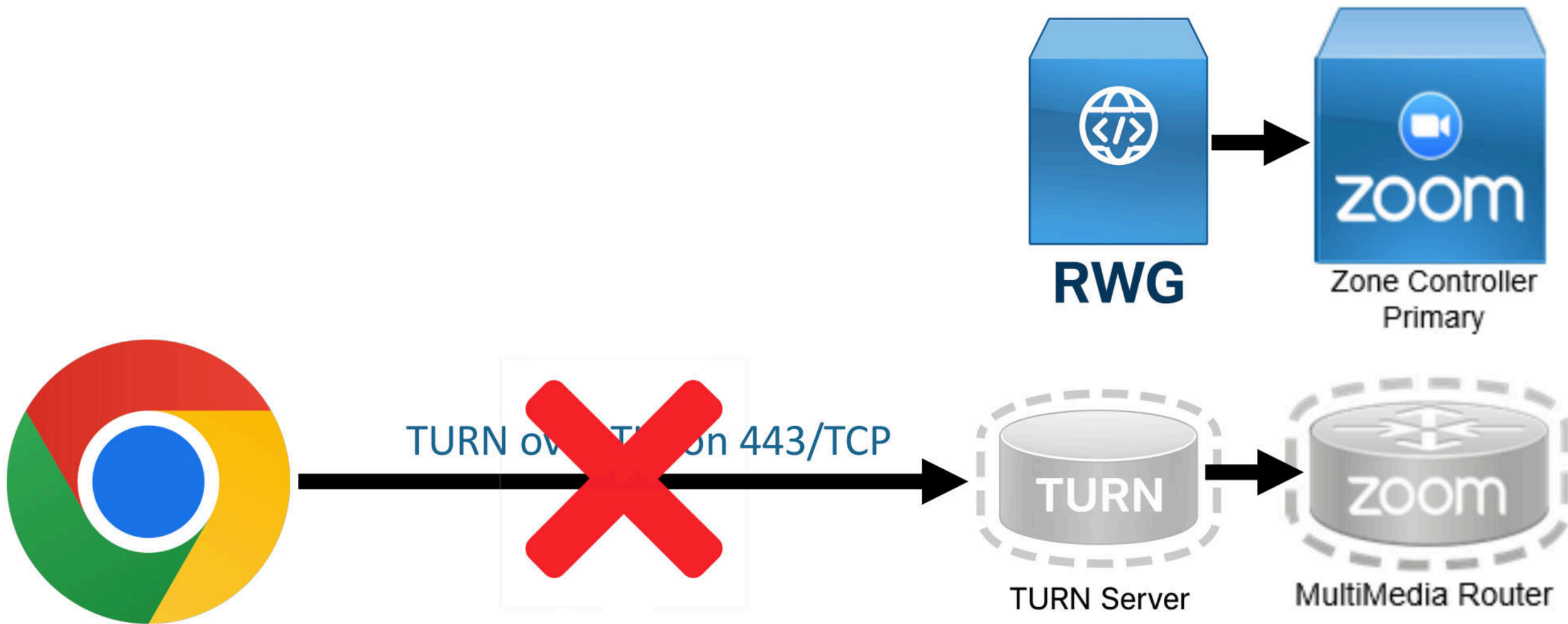
Example Zoom Web Client Egress Attempts



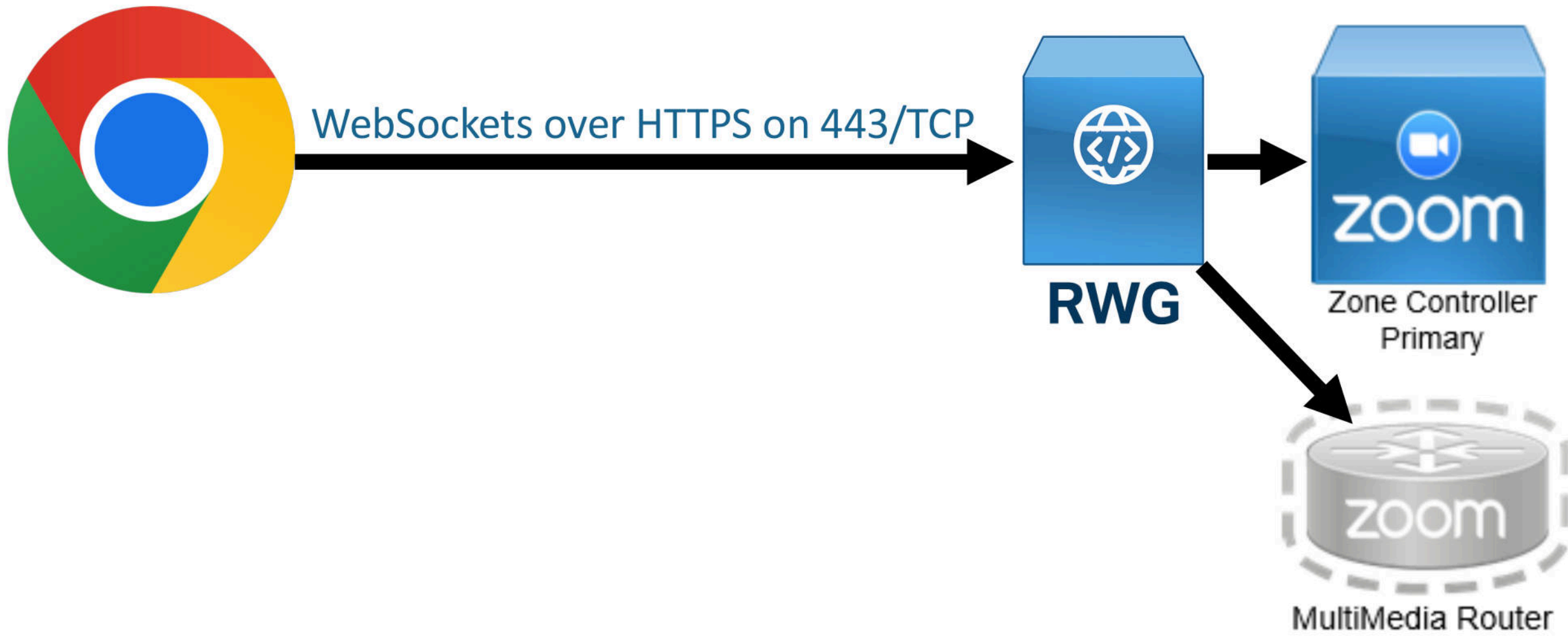
Example Zoom Web Client Egress Attempts



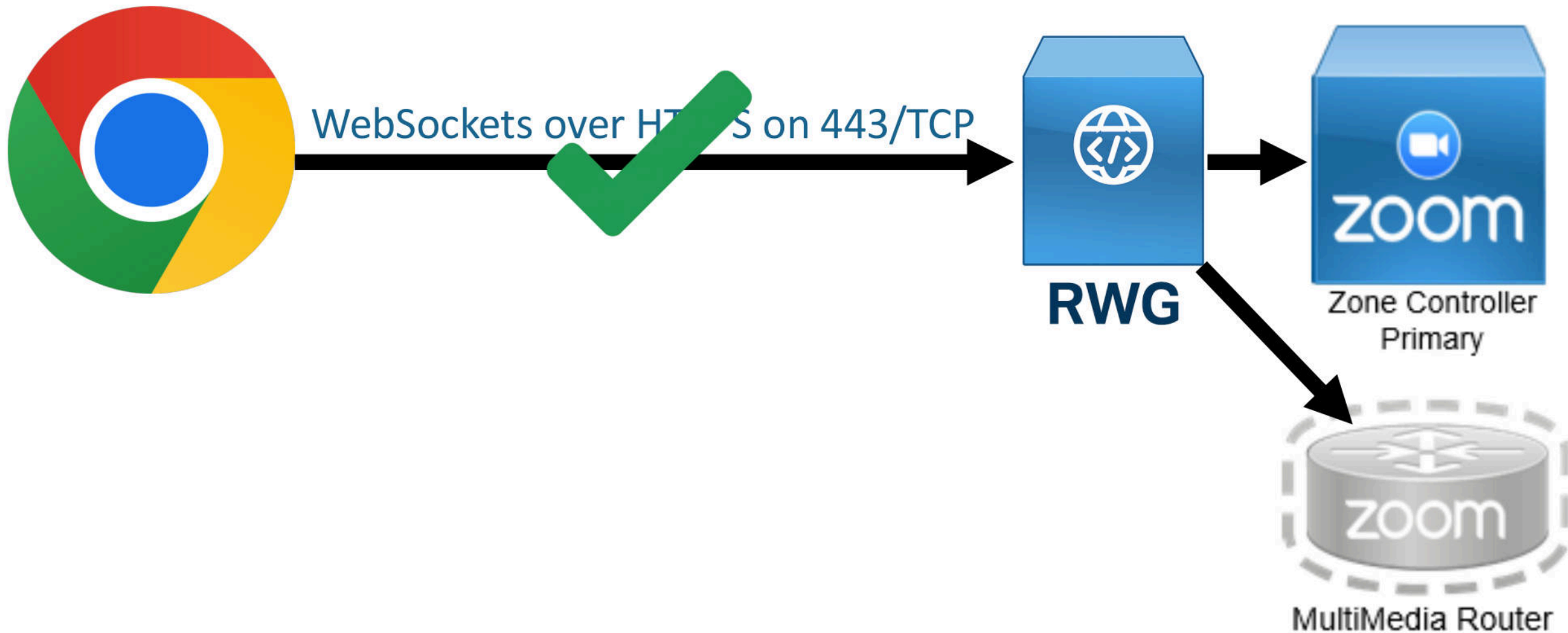
Example Zoom Web Client Egress Attempts



Example Zoom Web Client Egress Attempts



Example Zoom Web Client Egress Attempts



Highly Adaptable to Changing Environments



Developing the Capability

Analyzing Vendor Market Share

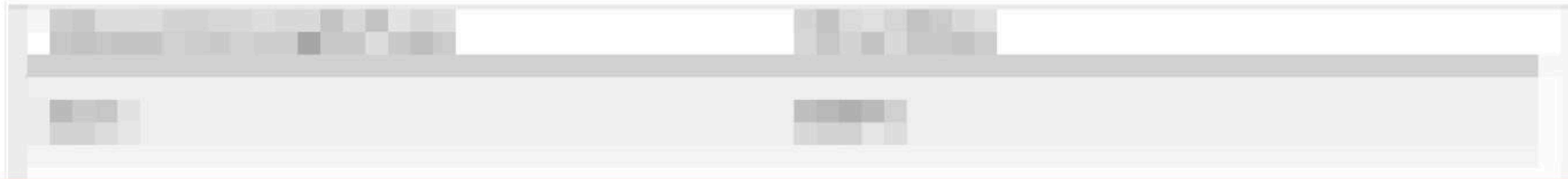
Videoconferencing Software	Market Share
Zoom	55.91%
Microsoft Teams	32.29%
GoToMeeting	8.81%
Google Meet	5.52%
WebEx	7.61%
RingCentral	5.31%
FaceTime	2.16%
Skype	1.41%
Facebook Messenger	0.75%
Bluejeans	0.31%

Source: [Statista](#)

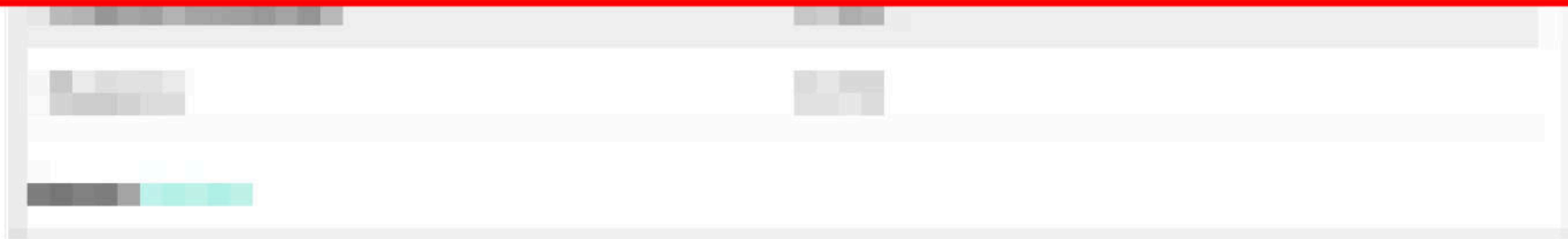
<https://www.demandsage.com/microsoft-teams-statistics/>



Analyzing Vendor Market Share



Videoconferencing Software	Market Share
Zoom	55.91%
Microsoft Teams	32.29%
GoToMeeting	8.81%
Google Meet	5.52%



<https://www.demandsage.com/microsoft-teams-statistics/>

Reverse Engineering Zoom

```
{
  "body":{
    "ABtoken":"3C45E3C9-7F73-2CD4-0C2A-61B0665E2AA7",
    "conID":"81423846-9F1F-D9EF-72D7-1265B18A9BBA",
    "confID":"0C71C7D6-C040-4363-94C1-3175DA4475F7",
    "e2eEncrypt":true,
    "elapsed":0,
    "encType":2,
    "hugeB0":true,
    "mediasdkConfig":{
      "iceServers":[
        {
          "credential":"rlYnbcRe9d5IqRiU/Ukst9QY0C2lidMWRmUQoWVvFoc=",
          "urls":"turns:turnsg02.cloud.zoom.us:443?transport=tcp",
          "username":"81423846-9F1F-D9EF-72D7-1265B18A9BBA:1741859664289"
        },
        {
          "credential":"y7rK3BSihbZ33NqeVtUsgynrdvJZpYRkUuukI6LaUpU=",
          "urls":"turns:turnsg01.cloud.zoom.us:443?transport=tcp",
          "username":"81423846-9F1F-D9EF-72D7-1265B18A9BBA:1741859664289"
        }
      ]
    },
    "meetingTopic":"Y29sYnkuZWxvdGVzdEBnbWFpbC5jb20ncyBab29tIE1lZXRpbmci",
    "mmrFeature":3204447728,
    "mmrFeatureEx":4501601879980014,
    "mmrFeatureExStr":"4616187620307367918",
    "mn":"97774758416",
    "participantID":238757,
    "participantIDStr":"238757",
    "reportDomain":"zoomsg134224146206rwg.cloud.zoom.us",
```


Reverse Engineering Zoom

```
"mediasdkConfig":{  
  "iceServers": [  
    {  
      "credential": "rlYnbcRe9d5IqRiU/Ukst9QY0C2lidMWRmUQoWVvFoc=",  
      "urls": "turns:turnsg02.cloud.zoom.us:443?transport=tcp",  
      "username": "81423846-9F1F-D9EF-72D7-1265B18A9BBA:1741859664289"  
    },  
    {  
      "credential": "y7rK3BSihbZ33NQeVtUsgynrdvJZpYRkUuukI6LaUpU=",  
      "urls": "turns:turnsg01.cloud.zoom.us:443?transport=tcp",  
      "username": "81423846-9F1F-D9EF-72D7-1265B18A9BBA:1741859664289"  
    }  
  ]  
},
```

Zooming in on TURN

Firewall rules for Zoom Meetings and Webinars

Protocol	Ports	Source	Destination
→ ~ nslookup	turnsg02.cloud.zoom.us		115.110.154.192/26 115.114.56.192/26 115.114.115.0/26 115.114.131.0/26 120.29.148.0/24 121.244.146.0/27 134.224.0.0/16 137.66.128.0/17 144.195.0.0/16 147.124.96.0/19 149.137.0.0/17 156.45.0.0/17 159.124.0.0/16 160.1.56.128/25 161.199.136.0/22 162.12.232.0/22 162.255.36.0/22 165.254.88.0/23
Server:	192.168.1.1		
Address:	192.168.1.1#53		
Non-authoritative answer:			
Name:	turnsg02.cloud.zoom.us		
Address:	134.224.147.10		
		source	

Reverse Engineering Microsoft Teams

Response

Pretty

Raw

Hex

Render



\n



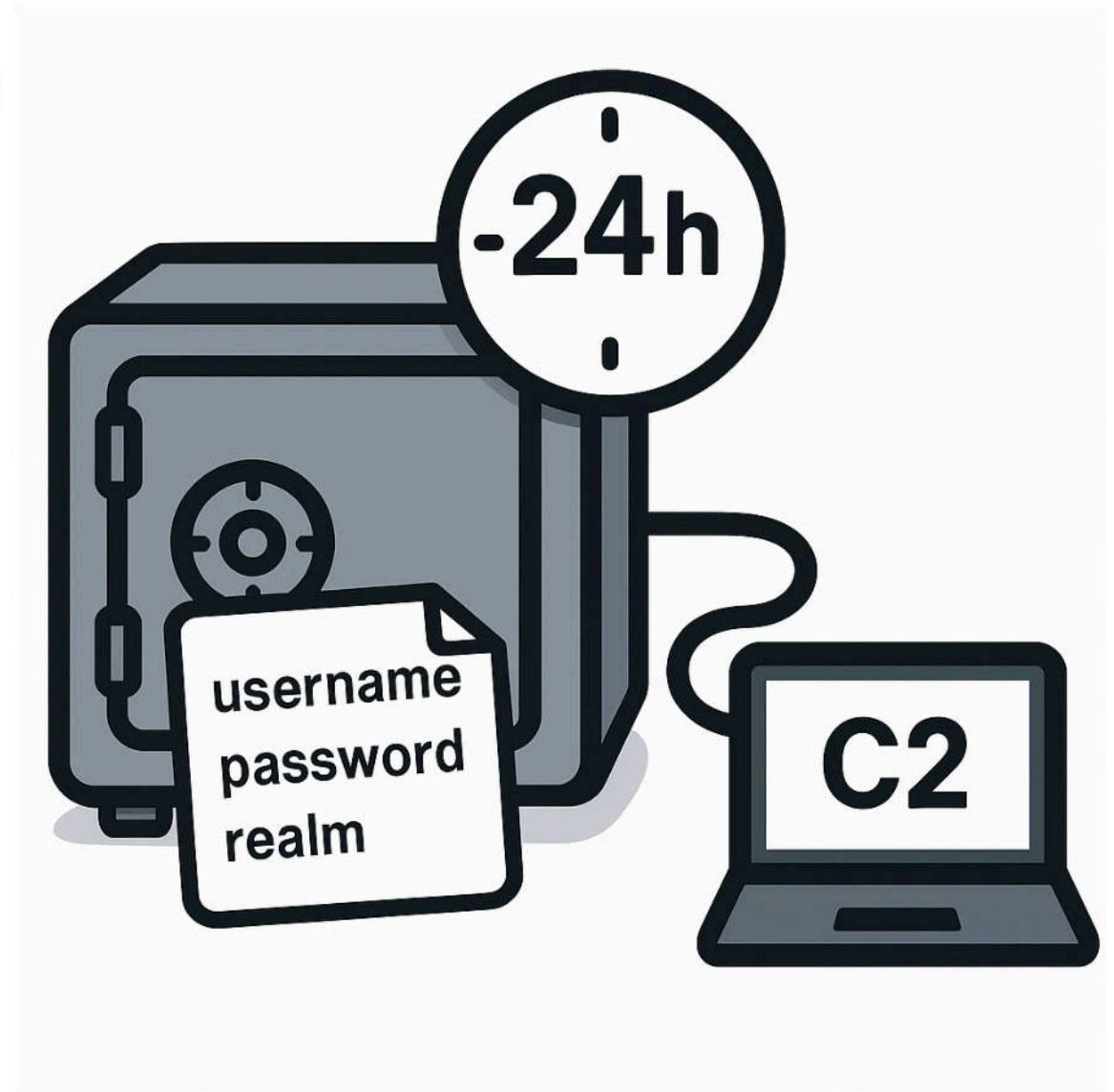
```
1 HTTP/2 200 OK
2 Cache-Control: no-cache, no-store
3 Content-Length: 186
4 Content-Type: application/json; charset=utf-8
5 Ms-Cv: lHKzC4UNR0280SVyWrqm+g.0
6 Strict-Transport-Security: max-age=31536000; includeSubDomains
7 Api-Supported-Versions: 1.0, 2.0
8 Server-Timing: reqlatency;dur=2
9 X-Cache: CONFIG_NOCACHE
10 X-MSedge-Ref: Ref A: C58D8123B9144A1CA7727C9B8DCD5BC8 Ref B: BKK30EDGE0511 Ref C: 2025-03-18T14:10:21Z
11 Date: Tue, 18 Mar 2025 14:10:21 GMT
12
13 {
  "tokens": [
    {
      "realm": "\"rtcmedia\"",
      "username": "AgAAJKTmRIAB252Px+6sqQkexkR0PUDm73PwpkvWP3IAAAAdg8G7t0lFUabmHtcNl06RCa80uA=",
      "password": "InNEjcnomvcT0pPEgPsA800mMkE="
    }
  ],
  "expires": 604800
}
```

Reverse Engineering Microsoft Teams

```
{
  "tokens": [
    {
      "realm": "\"rtcmedia\"",
      "username": "AgAAJKTmRIAB252Px+6sqQkexkR0PUDm73PwpkvWP3IAAAAAAg8G7t0lFUabmHtcNl06RCa80uA=",
      "password": "InNEjcnomvcT0pPEgPsA800mMkE="
    }
  ],
  "expires": 604800
}
```


Observations Regarding TURN Credentials

- Usually valid for a couple of days
- Complements an existing long-term channel
- Not tied to specific calls and credentials persist post-session
- Applies to common platforms like Zoom and Teams
- No install or meeting required on the victim side





What do we want to build?

- A short-lived tunnel launched from an existing implant
- Used briefly and mimics activity like a video-call
- Runs in parallel with long-term infrastructure
- Lightweight enough to avoid clogging that primary channel
- Disguised among high-traffic destinations (e.g., Zoom, Teams)

Most Visited Sites

Domain	Request Volume
• google.com	82.450
• microsoft.com	67.813
• amazon.com	52.209
• somebank.com	47.652

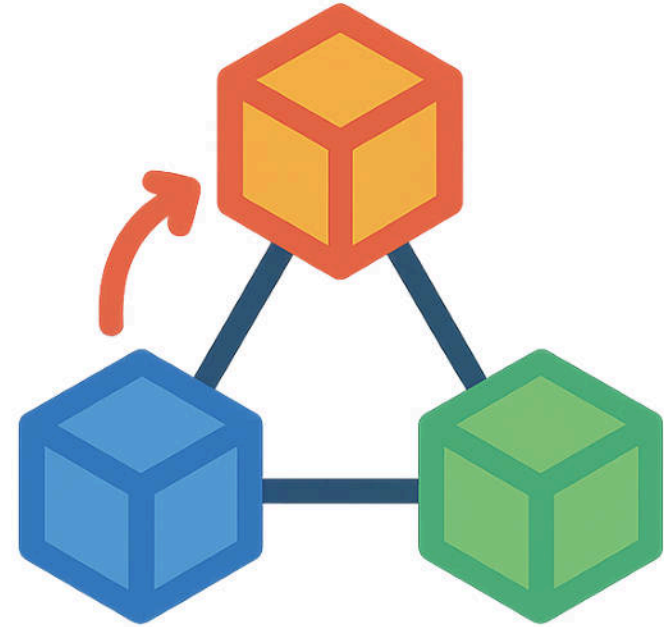
TURNt (TURN tunneler): A Tool for Short-Term C2



<https://github.com/praetorian-inc/turnt>

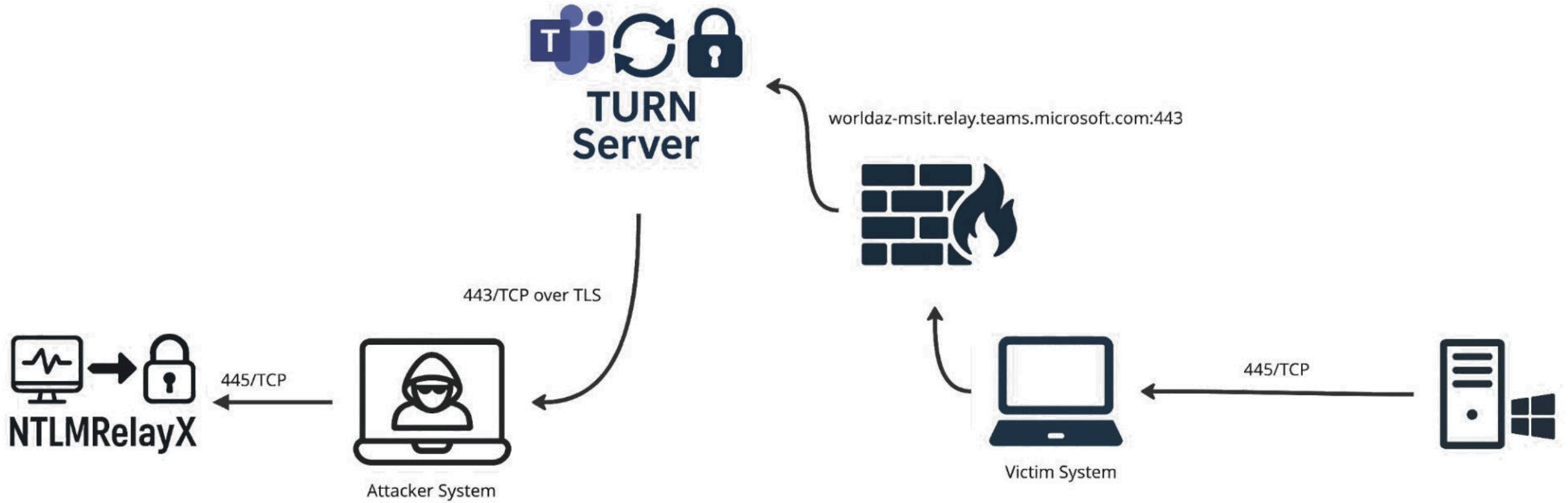
Use-Cases and Capabilities

- Fast tunnel setup during assumed breach scenarios
- No need to provision infrastructure in advance
- Operates from operator laptop or disposable VDI
- Ideal for decentralized red team operations
- Lightweight, flexible, and serverless by design

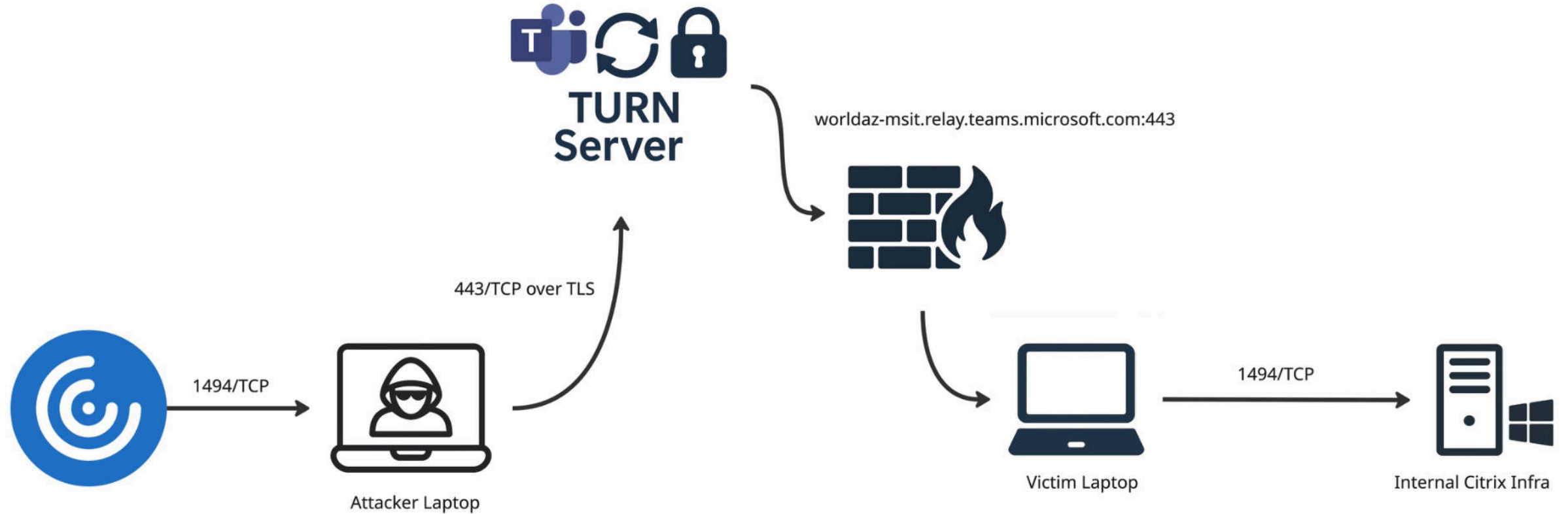


Decentralized C2

Remote Port Forwarding



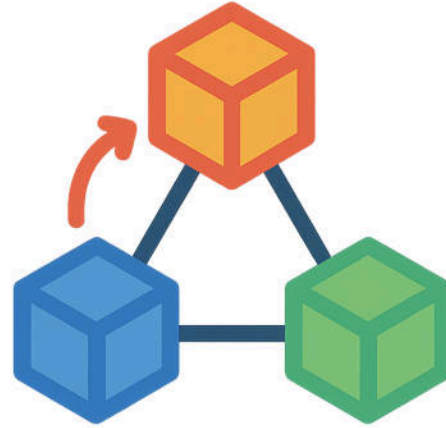
Local Port Forwarding



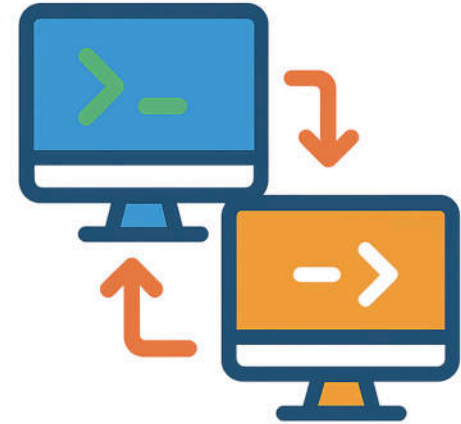
Use-Cases and Capabilities



**SOCKS
Proxying**



**Decentralized
C2**



**Local & Remote
Port-Forwarding**

Zoom Demo Example Scenario

- Obtaining credentials from Zoom
- Victim doesn't need to do anything
- Laptop is the operator laptop
- Example victim system is GCP virtual machine
- Demo downloading a file through the channel



**TURNt
Controller**



**TURNt
Relay**

Zoom Video Demo

[demo video](#)

Examining Wireshark Traffic (Zoom)

- > Internet Protocol Version 4, Src: 192.168.1.43, Dst: 170.114.166.217
- > Transmission Control Protocol, Src Port: 61862, Dst Port: 443, Seq: 1, Ack: 1, Len: 273
- ✓ Transport Layer Security
 - ✓ TLSv1.3 Record Layer: Handshake Protocol: Client Hello
 - Content Type: Handshake (22)
 - Version: TLS 1.0 (0x0301)
 - Length: 268
 - ✓ Handshake Protocol: Client Hello
 - Handshake Type: Client Hello (1)
 - Length: 264
 - > Version: TLS 1.2 (0x0303)
 - Random: cfda9068dcfcc75da6d6220358f91edb332335072707b911d42c681f029daa1f
 - Session ID Length: 32
 - Session ID: fa209d4bc07da86474eca647841334608eab1b481800e7e41def9d2e02d184f
 - Cipher Suites Length: 38
 - > Cipher Suites (19 suites)
 - Compression Methods Length: 1
 - > Compression Methods (1 method)
 - Extensions Length: 153
 - > Extension: server_name (len=26) name=turnsin01.sin.zoom.us

Examining Wireshark Traffic (Zoom)

▼ Handshake Protocol: Client Hello

Handshake Type: Client Hello (1)

Length: 264

> Version: TLS 1.2 (0x0303)

Random: cfda9068dcfcc75da6d6220358f91edb332335072707b911d42c681f029daa1f

Session ID Length: 32

Session ID: fa209d4bc07da86474eca647841334608eab1b481800e7e41deff9d2e02d184f

Cipher Suites Length: 38

> Cipher Suites (19 suites)

Compression Methods Length: 1

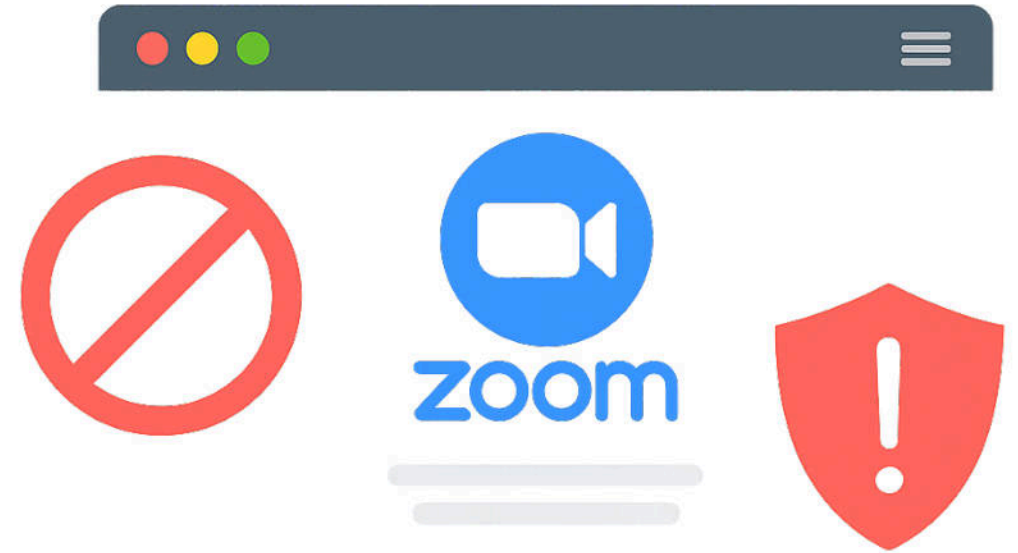
> Compression Methods (1 method)

Extensions Length: 153

> Extension: server_name (len=26) name=turnsin01.sin.zoom.us

Zoom Recent Mitigations

- Pushed out a mitigation on Sunday
- Prevents peer-to-peer connections via TURN infra
- Restricted to client and media server ranges
- Haven't dug into potential bypasses
- Feasibility depends on the provider

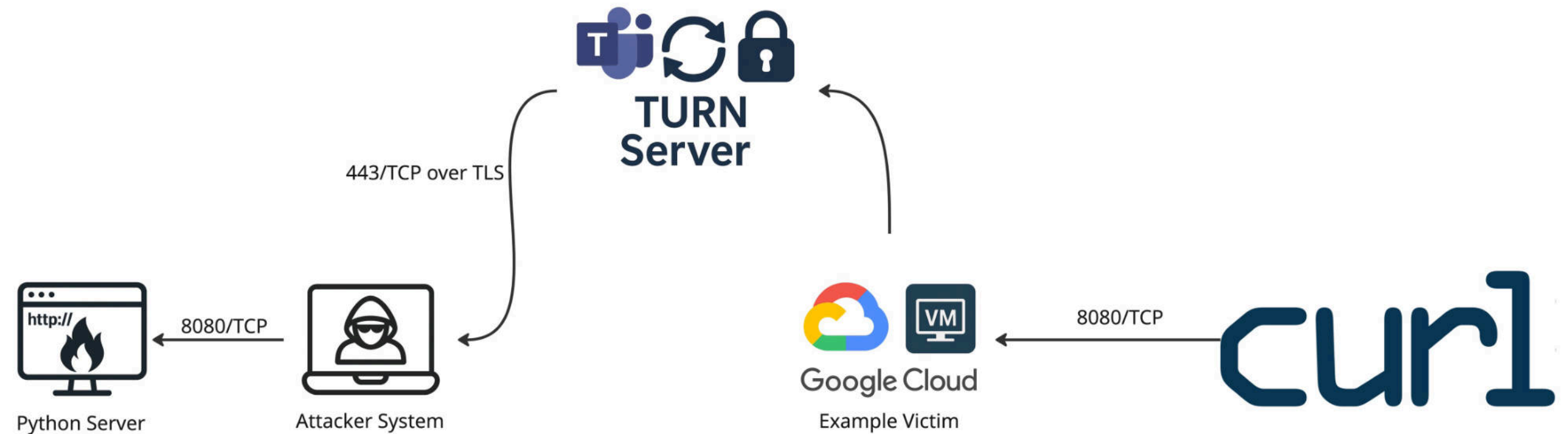


Microsoft Teams Demo

- Show automated retrieval of TURN credentials from Microsoft
- Demonstrate a speed test showing a 100 MB file download
- Demonstrate remote port-forwarding capability
- Lab uses my local laptop and a demo virtual machine in GCP



Microsoft Teams Demo



Microsoft Teams Video Demo

[demo video](#)

Examining Wireshark Traffic (Microsoft Teams)

- > Internet Protocol Version 4, Src: 192.168.1.43, Dst: 52.114.55.197
- > Transmission Control Protocol, Src Port: 60570, Dst Port: 443, Seq: 1429, Ack: 1, Len: 357
- > [2 Reassembled TCP Segments (1785 bytes): #30829(1428), #30830(357)]

✓ Transport Layer Security

✓ TLSv1.2 Record Layer: Handshake Protocol: Client Hello

Content Type: Handshake (22)

Version: TLS 1.0 (0x0301)

Length: 1780

✓ Handshake Protocol: Client Hello

Handshake Type: Client Hello (1)

Length: 1776

> Version: TLS 1.2 (0x0303)

> Random: 21058fee53f9753786f537e6158e9f9123d3ce824bfff6da2a14f47c80a2e9b00

Session ID Length: 32

Session ID: 3b39951ffe287758fdf1c8ca54b945f99fc5ae94a7891877b7652a9e148230b5

Cipher Suites Length: 32

> Cipher Suites (16 suites)

Compression Methods Length: 1

> Compression Methods (1 method)

Extensions Length: 1671

> Extension: Reserved (GREASE) (len=0)

> Extension: status_request (len=5)

> Extension: signature_algorithms (len=18)

> Extension: key_share (len=1263) X25519MLKEM768, x25519

> Extension: encrypted_client_hello (len=250)

> Extension: server_name (len=43) name=worldaz-msit.relay.teams.microsoft.com

Examining Wireshark Traffic (Microsoft Teams)



- > Internet Protocol Version 4, Src: 192.168.1.43, Dst: 52.114.55.197
- > Transmission Control Protocol, Src Port: 60570, Dst Port: 443, Seq: 1429, Ack: 1, Len: 357
- > [2 Reassembled TCP Segments (1785 bytes): #30829(1428), #30830(357)]
- ✓ Transport Layer Security
 - ✓ TLSv1.2 Record Layer: Handshake Protocol: Client Hello
 - Content Type: Handshake (22)
 - Version: TLS 1.0 (0x0301)
 - Length: 1780
 - ✓ Handshake Protocol: Client Hello



Examining Wireshark Traffic (Microsoft Teams)

- ✓ Handshake Protocol: Client Hello
 - Handshake Type: Client Hello (1)
 - Length: 1776
 - > Version: TLS 1.2 (0x0303)
 - > Random: 21058fee53f9753786f537e6158e9f9123d3ce824bff6da2a14f47c80a2e9b00
 - Session ID Length: 32
 - Session ID: 3b39951ffe287758fdf1c8ca54b945f99fc5ae94a7891877b7652a9e148230b5
 - Cipher Suites Length: 32
 - > Cipher Suites (16 suites)
 - Compression Methods Length: 1
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 - Extensions Length: 1671
 - > Extension: Reserved (GREASE) (len=0)
 - > Extension: status_request (len=5)
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 - > Extension: encrypted_client_hello (len=250)
 - > Extension: server_name (len=43) name=worldaz-msit.relay.teams.microsoft.com



BGP

→ BGP

...

→

...

→

TIER 2 ISP

TIER 3 ISP

IXP

DATA
CENTER



DISCLAIMER

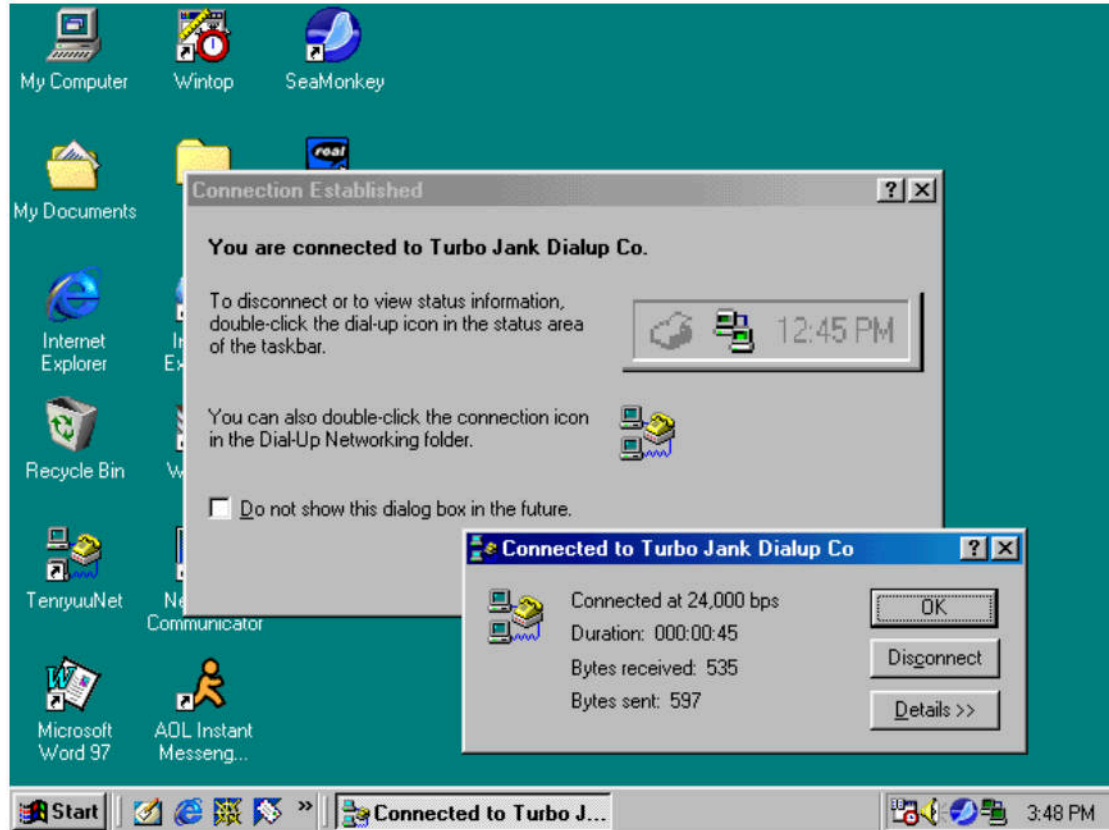
TIER 3 ISP

Compromising the Helpdesk

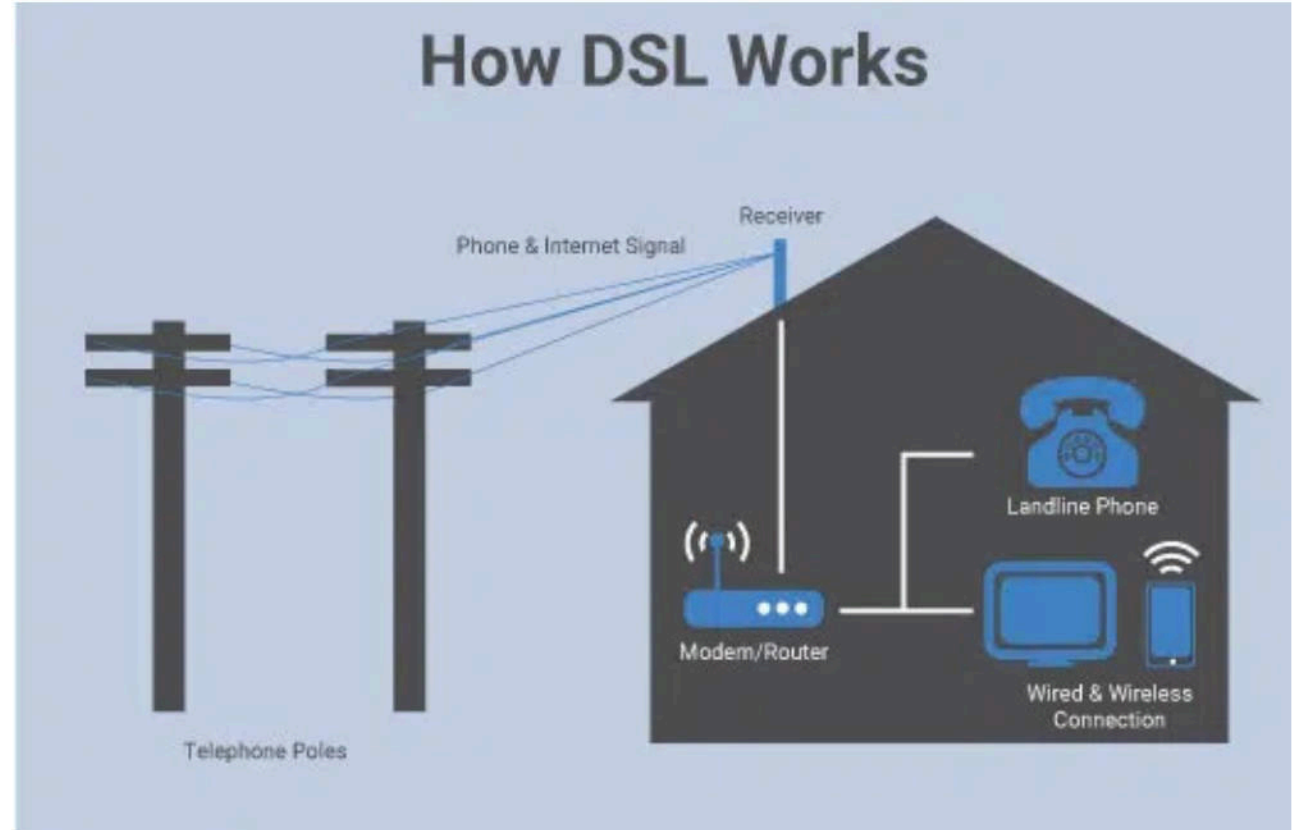
- Helpdesk accounts are often privileged easy targets
- Common red team foothold especially in large orgs
- Frequently outsourced overseas locations
- Ran into slow unreliable tunnel from compromised users
- Example: Slow tunneling through helpdesk employee in India



Last Mile Delivery is the Traditional Bottleneck



[source](#)



[source](#)

Fiber to the Home Build Out

Best **1 Gbps** broadband plans for you

₹50 / Month, one-time benefit

 **₹ 4200** / Month

X 3 Months = ₹ 12600 + Gst

 Upload/Download = **1 Gbps**
Unlimited Data, Unstoppable Speed

 **Enjoy Entertainment Bundle**

OTTs - ALL

Channels - 200+



More Details

[Get Started](#)

₹50 / Month, one-time benefit

 ~~₹4200~~
₹ 3900 / Month

X 6 Months = ₹ 23400 + Gst

 Upload/Download = **1 Gbps**
Unlimited Data, Unstoppable Speed

 **Enjoy Entertainment Bundle**

OTTs - ALL

Channels - 200+



Free Wi-Fi 6 router

More Details

[Get Started](#)

₹50 / Month, one-time benefit

 ~~₹4200~~
₹ 3600 / Month

X 12 Months = ₹ 43200 + Gst

 Upload/Download = **1 Gbps**
Unlimited Data, Unstoppable Speed

 **Enjoy Entertainment Bundle**

OTTs - ALL

Channels - 200+

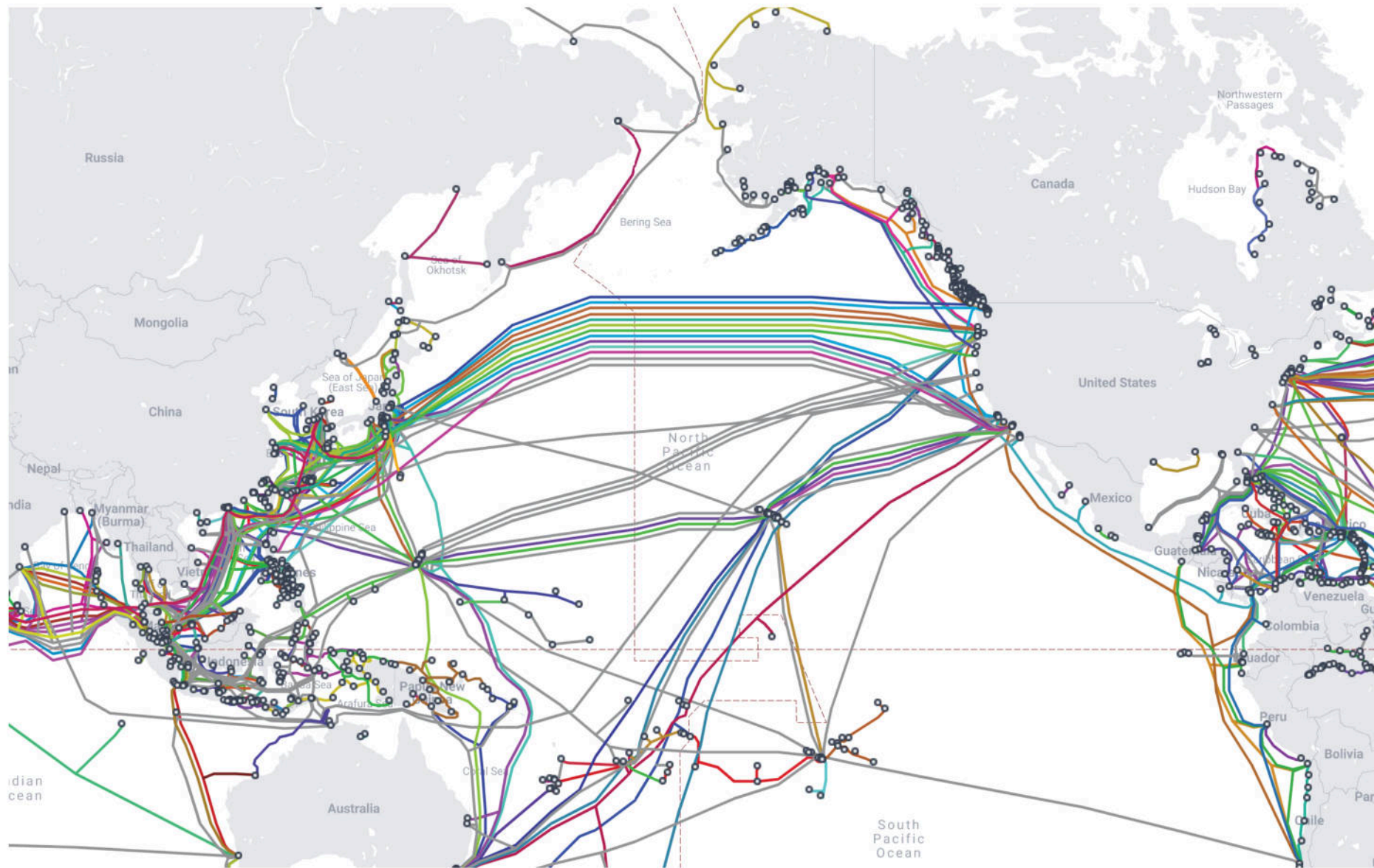


Free Wi-Fi 6 + Mesh router

More Details

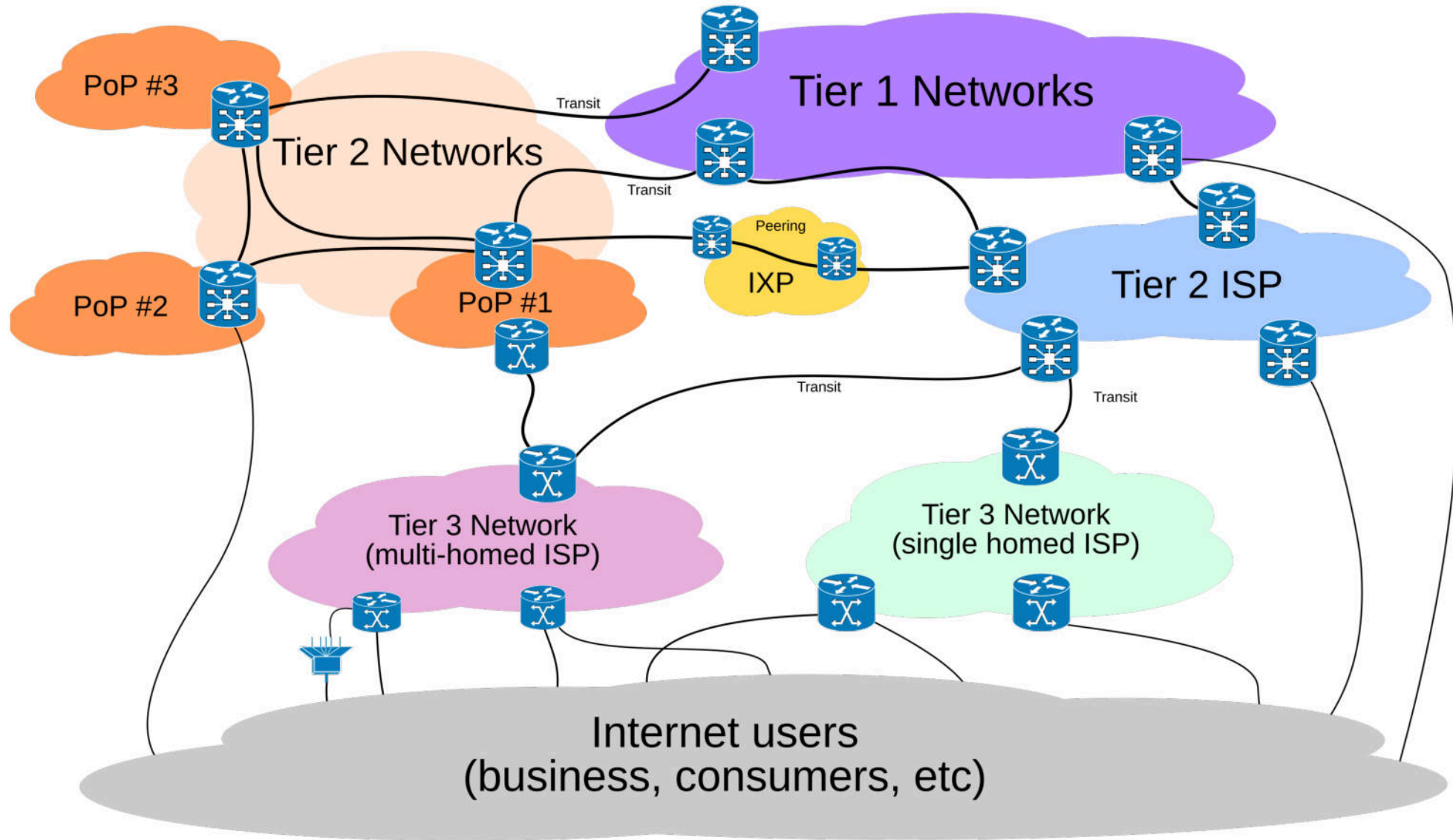
[Get Started](#)

Global Fiber Connectivity Map

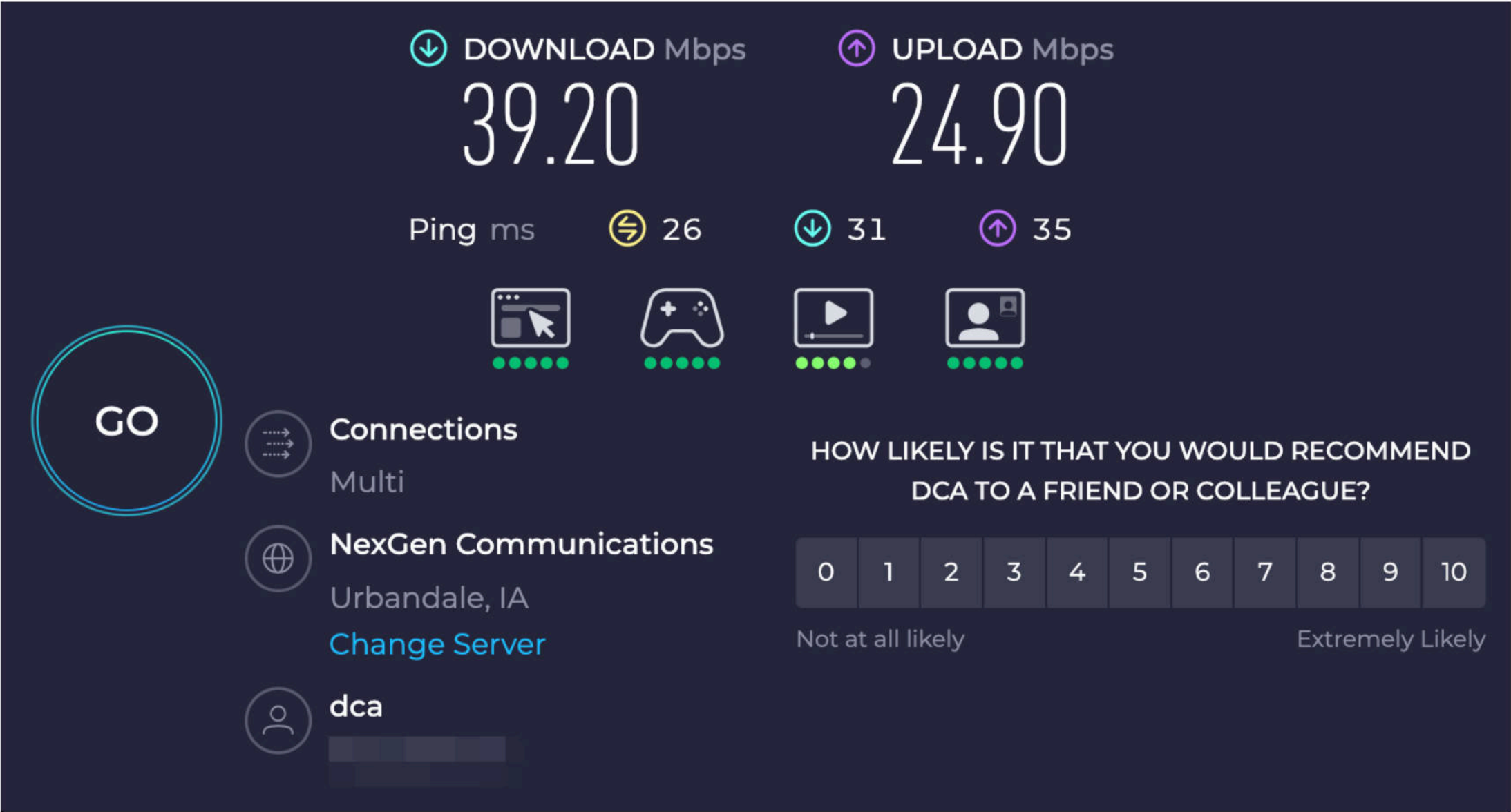


[source](#)

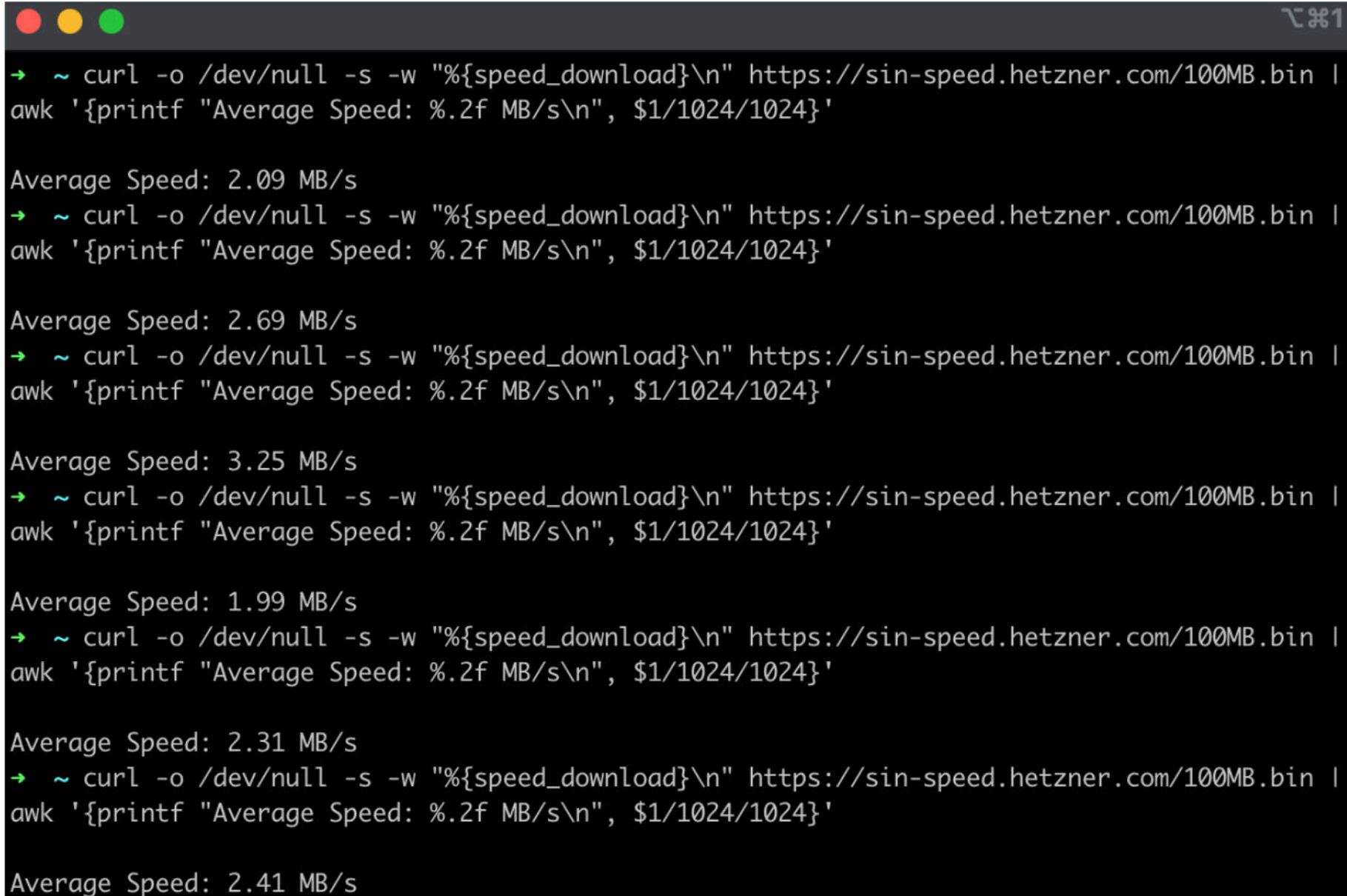
Internet Routing is a Lot Messier Than Expected



Routing Can Impact Performance Significantly (ExpressVPN)



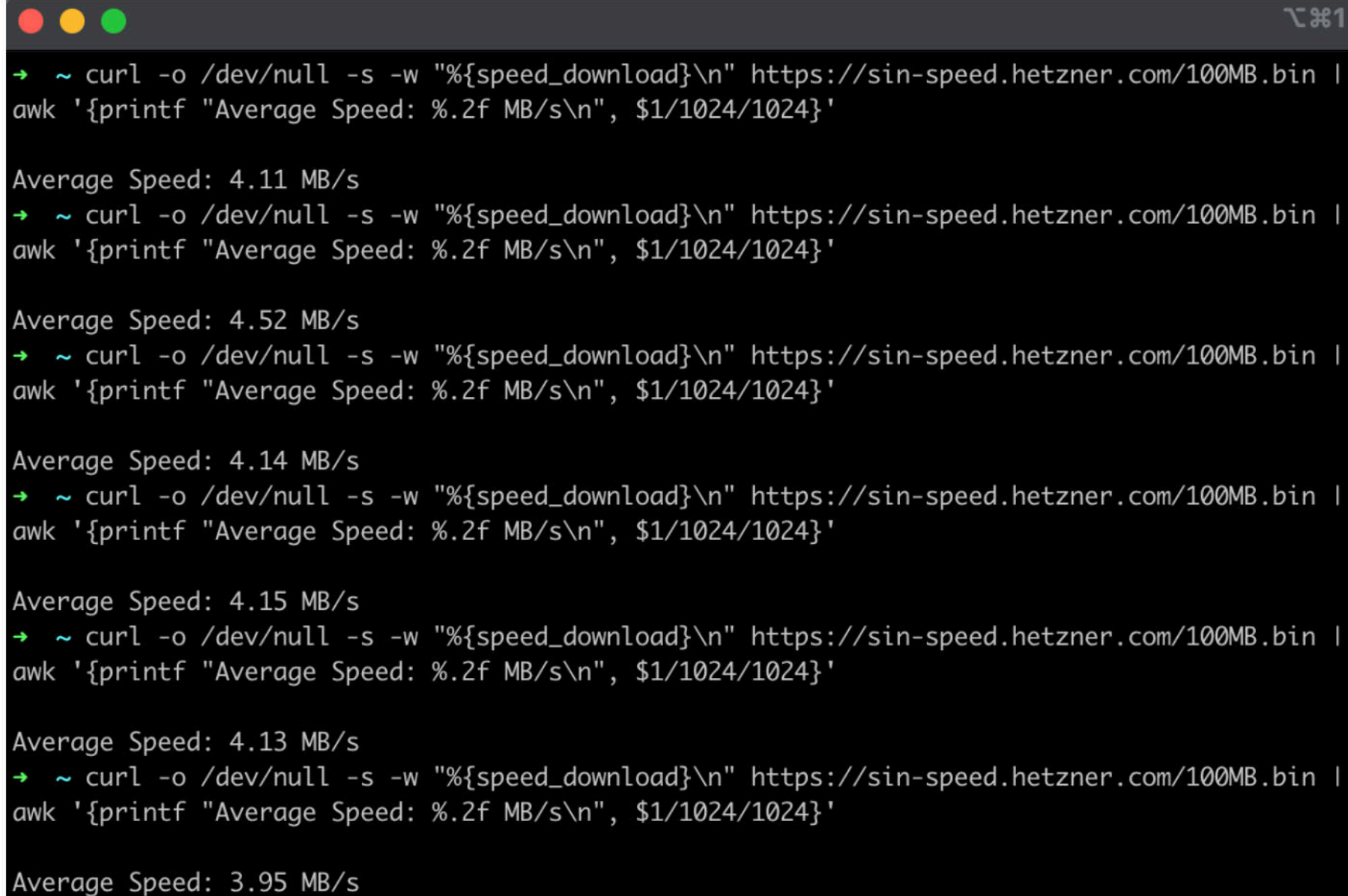
Routing Can Impact Performance Significantly (ExpressVPN)



A terminal window with a dark background and light-colored text. The window has three colored window control buttons (red, yellow, green) in the top-left corner and a title bar with the text "Terminal" in the top-right corner. The terminal displays a series of commands and their outputs, showing download speeds from a specific server.

```
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 2.09 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 2.69 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 3.25 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 1.99 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 2.31 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 2.41 MB/s
```

Routing Can Impact Performance Significantly (ExpressVPN)

A terminal window with a dark background and light-colored text. The window has three colored window control buttons (red, yellow, green) in the top-left corner and a title bar with the text "\#1" in the top-right corner. The terminal displays a series of commands and their outputs, showing network speed tests. The commands use curl to download a file from https://sin-speed.hetzner.com/100MB.bin and awk to format the output. The outputs show average speeds in MB/s.

```
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 4.11 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 4.52 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 4.14 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 4.15 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 4.13 MB/s  
→ ~ curl -o /dev/null -s -w "%{speed_download}\n" https://sin-speed.hetzner.com/100MB.bin |  
awk '{printf "Average Speed: %.2f MB/s\n", $1/1024/1024}'  
  
Average Speed: 3.95 MB/s
```

Routing Can Impact Performance Significantly (ExpressVPN)

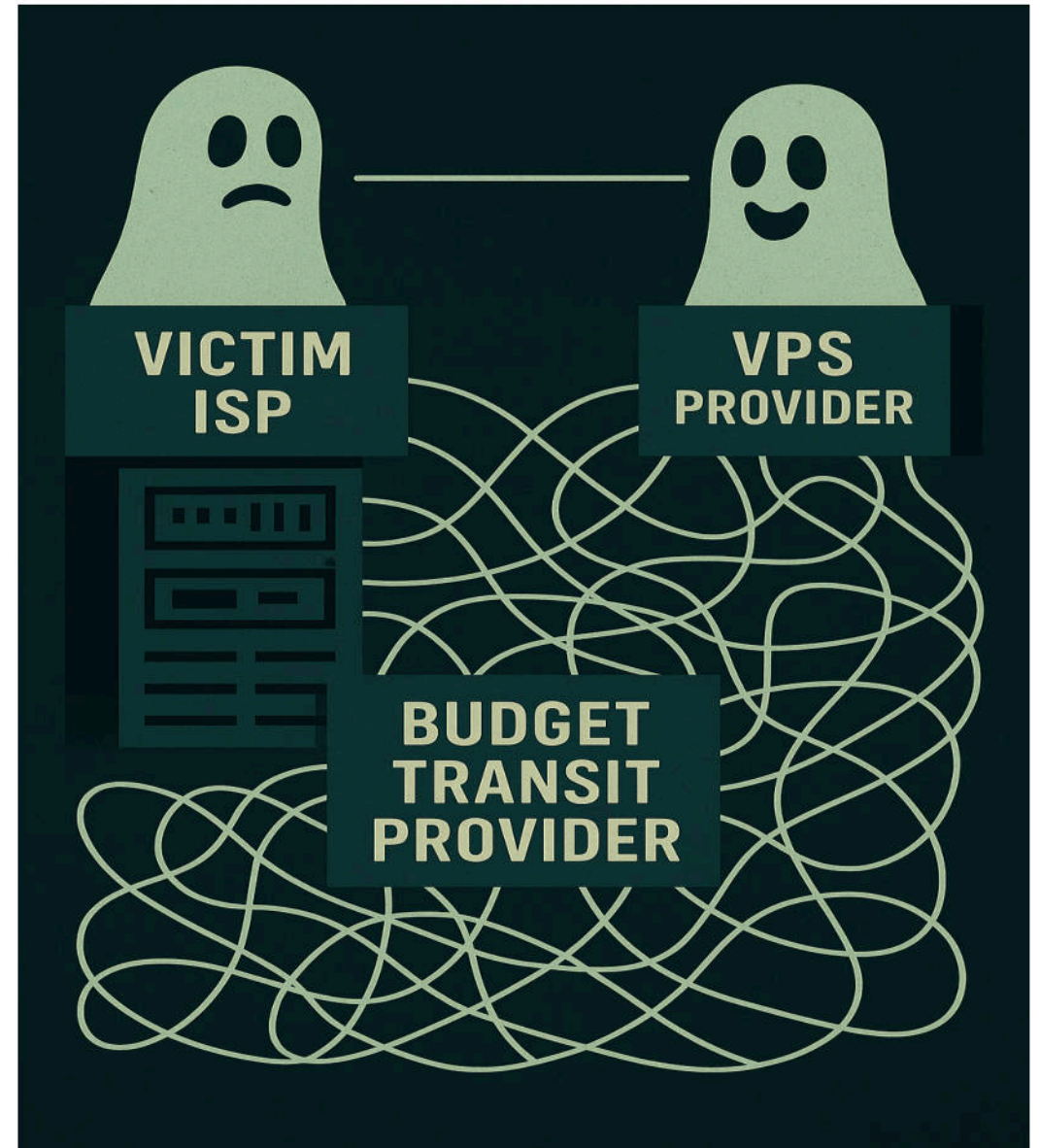
```
→ ~ traceroute 5.223.7.195
traceroute to 5.223.7.195 (5.223.7.195), 64 hops max, 40 byte packets
 1  192.168.8.1 (192.168.8.1)  2.847 ms  2.333 ms  1.637 ms
 2  10.180.6.65 (10.180.6.65)  7.319 ms  9.886 ms  7.806 ms
 3  10.180.6.163 (10.180.6.163)  20.904 ms  33.884 ms  65.594 ms
 4  10.9.180.3 (10.9.180.3)  24.578 ms  14.115 ms  14.705 ms
 5  10.198.9.4 (10.198.9.4)  16.509 ms  14.164 ms  15.035 ms
 6  ln-oix-gw.edge.dcaonline.com (10.6.9.1)  17.582 ms  20.969 ms  16.844 ms
 7  * * 206.126.235.20 (206.126.235.20)  22.966 ms
 8  * * *
 9  port-channel10.core2.slc1.he.net (72.52.92.42)  42.848 ms * *
10  * * *
11  100ge0-73.core4.lax2.he.net (184.105.222.113)  57.986 ms * *
12  * port-channel6.core3.tyo1.he.net (184.105.213.118)  151.090 ms *
13  * * *
14  viewqwest-pte-ltd.port-channel2.switch2.sin1.he.net (27.50.33.94)  285.859 ms  427.768 m
s  305.979 ms
15  132.147.112.108 (132.147.112.108)  307.782 ms  216.672 ms  219.747 ms
16  fnet118-f60-60-access.vqbn.com.sg (103.60.60.118)  226.629 ms  217.943 ms  216.752 ms
17  * * *
18  * * *
19  * * *
20  * |
```


Routing Can Impact Performance Significantly (ExpressVPN)

```
→ ~ traceroute 5.223.7.195
traceroute to 5.223.7.195 (5.223.7.195), 64 hops max, 40 byte packets
 1  * * *
 2  181.215.65.124 (181.215.65.124)  27.712 ms  39.643 ms  24.695 ms
 3  vl221.chi-cs1-core-1.cdn77.com (79.127.195.25)  26.475 ms
    vl221.chi-cs1-core-2.cdn77.com (79.127.195.26)  25.000 ms  23.367 ms
 4  ae-7.a04.chcgil11.us.bb.gin.ntt.net (128.241.10.28)  26.636 ms  25.495 ms  24.839 ms
 5  ae-14.r24.chcgil09.us.bb.gin.ntt.net (129.250.4.72)  25.125 ms  26.765 ms
    ae-13.r24.chcgil09.us.bb.gin.ntt.net (129.250.3.188)  25.447 ms
 6  * * *
 7  ae-6.r26.osakjp02.jp.bb.gin.ntt.net (129.250.3.61)  166.132 ms *  177.447 ms
 8  * * ae-0.r27.osakjp02.jp.bb.gin.ntt.net (129.250.3.45)  157.599 ms
 9  ae-7.r24.sngpsi07.sg.bb.gin.ntt.net (129.250.2.66)  766.735 ms  300.492 ms  804.168 ms
10  ae-15.a03.sngpsi07.sg.bb.gin.ntt.net (129.250.6.65)  273.625 ms  593.517 ms  339.151 ms
11  ae-0.hetzner.sngpsi07.sg.bb.gin.ntt.net (116.51.31.51)  993.378 ms  229.521 ms  381.745
ms
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
```

Peering Can Be More Important Than Uplink Speed

- Example: Imagine trying to exfil 10GB of data from a compromise webserver
- Good peering beats raw bandwidth for tunnel speed
- Exfil to servers with good peering with victim internet provider
- Budget hosting providers often cheap on transit
- Some region have fast local fiber but bad global peering
- This could be an entire talk



Performance Killers When Tunneling over TCP

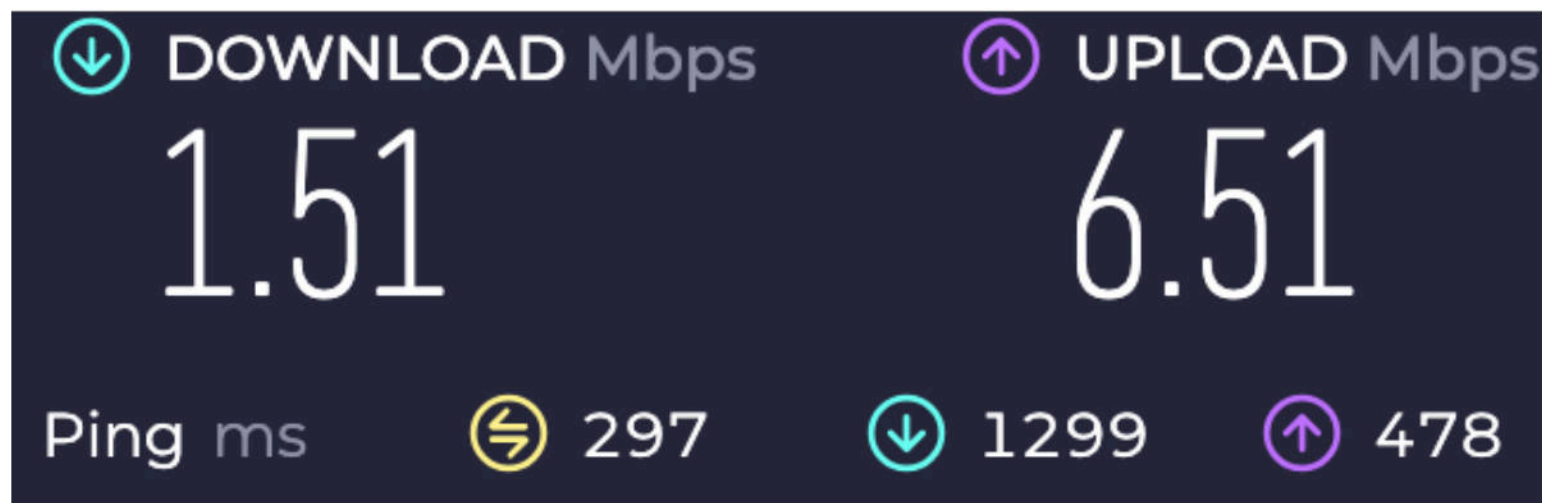
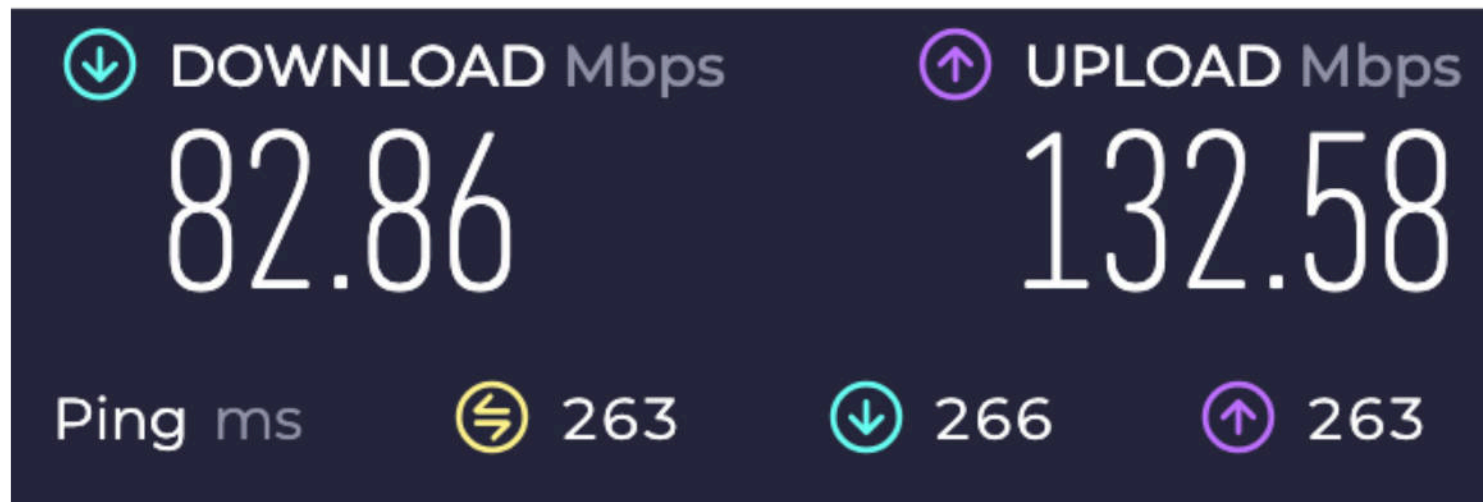


TCP Meltdown

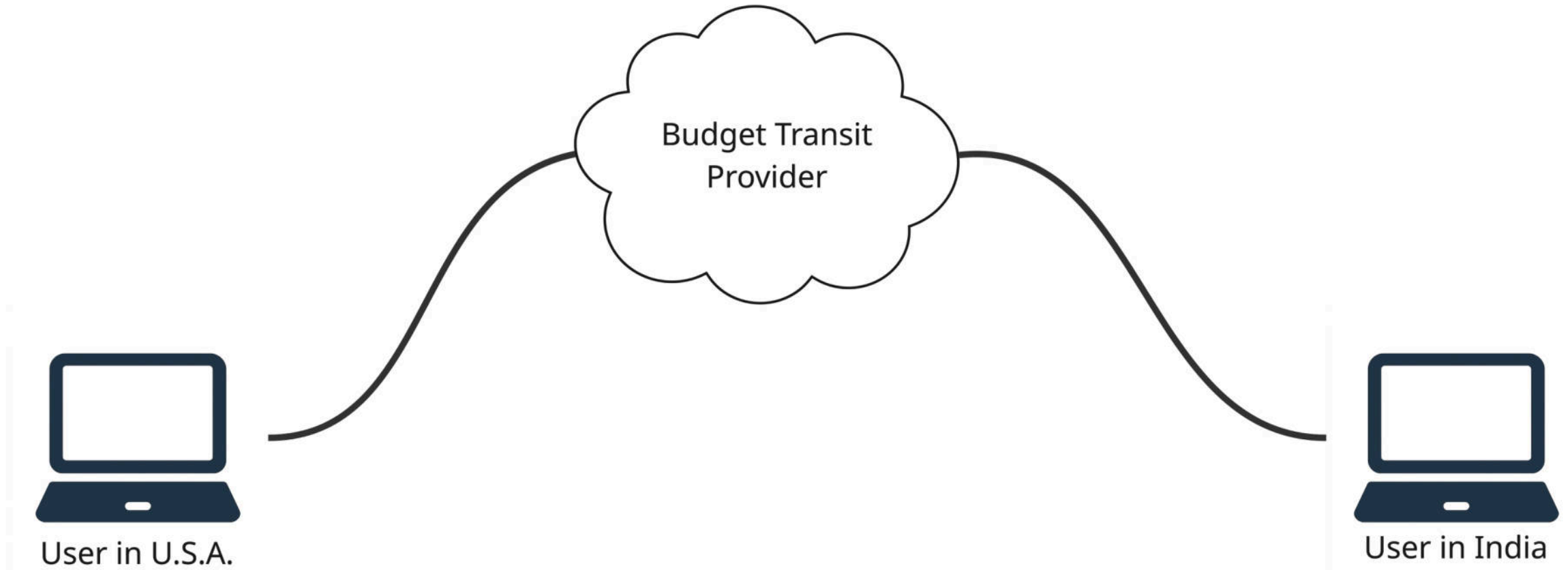


**Head-Of-Line
Blocking**

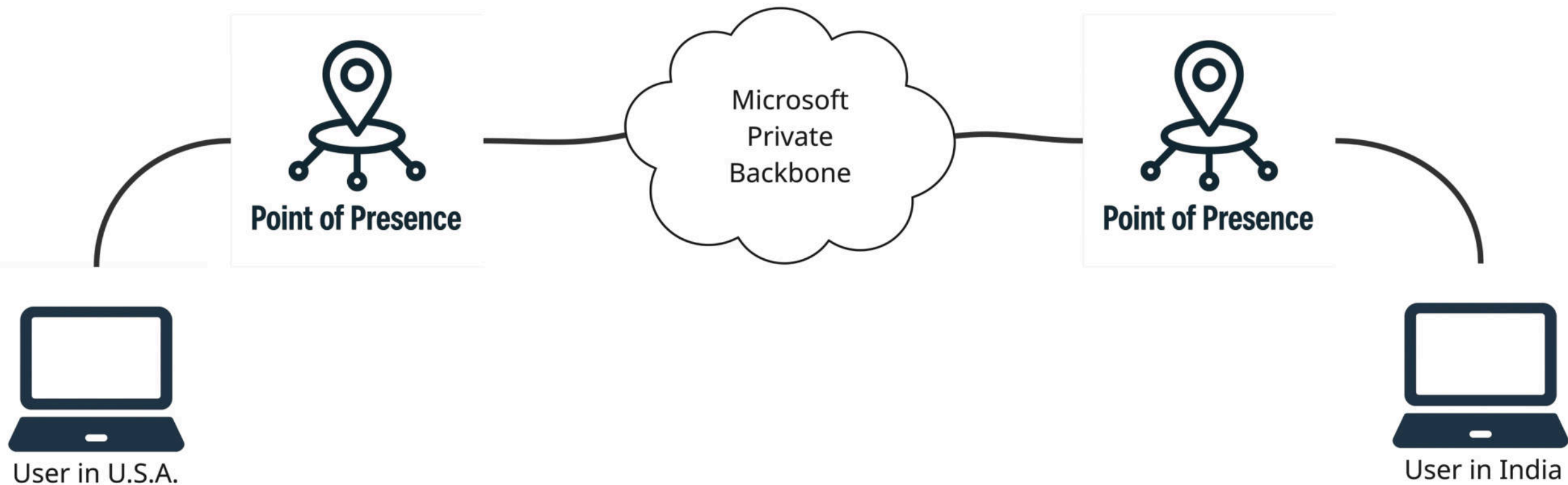
TCP Meltdown in Action



Peer to Peer Isn't Always the Most Efficient



Peer to Peer Isn't Always the Most Efficient



Recommend: Microsoft Teams for Cross Continent Tunneling





Questions?

Blog Post



Tool Release



LinkedIn



Blog Post: <https://www.praetorian.com/blog/ghost-calls-abusing-web-conferencing-for-covert-command-and-control-part-1/>

GitHub: <https://github.com/praetorian-inc/turnt>

LinkedIn: <https://www.linkedin.com/in/adam-crosser-366263265>

X: <https://x.com/UNC1739>